Fire and Punishment: Enforcement incentives and fire use in the Brazilian Amazon

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Incomplete monitoring of illegal behavior may cause violators to reoptimize their criminal activities. I study how satellite monitoring of deforestation impacts illegal land-clearing behavior in the Brazilian Amazon. I argue that by detecting primarily mechanical clearings, the introduction of near-real-time satellite monitoring favored a technological substitution towards the use of fire. Using cloud cover as an instrument for satellite-based detection, I find that municipalities under more intensive surveillance experienced a statistically significant substitution from mechanical to fire-based clearing. My findings help explain the persistence of Amazon fires despite the decline in deforestation starting from 2005.

Keywords: Forest fires, deforestation, environmental law enforcement

JEL Codes: Q51, Q57, Q58

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1 Introduction

> *Our sense is that people are betting on this idea: taking advantage of the fact that it is dry and hot, and the forest no longer has the same humidity as it used to. It's a window of opportunity: "Since I can no longer deforest, I'll set it on fire."*
>
> - Rodrigo Agostinho, *former IBAMA president, quoted at Girardi (2024)*

Forest fires impose severe negative externalities on the environment and human life (Wen et al., 2023; Heft-Neal et al., 2023; Pullabhotla et al., 2023). While fires start mostly unintentionally in high-income countries, intentional anthropogenic activity is the key driver in the tropics (Balboni et al., 2024). In this context, fire is used to clear forests and substitutes for mechanical clearing, which uses equipment such as chainsaws and excavators. Importantly, fire-based deforestation creates additional externalities due to the risk of spread and the smoke that affects areas beyond the cleared parcel. Therefore, policies that change the relative cost of the two methods change the social cost of each cleared hectare. Yet we have little evidence on how environmental policy affects deforestation technology choice.

In this paper, I study how environmental law enforcement affects deforestation behavior, distinguishing between mechanical clearing and fire-based clearing in the Brazilian Amazon. I explore the introduction of a near-real-time, satellite-based alert system called DETER, which is used to detect illegal deforestation and trigger on-the-ground enforcement. Crucially, DETER has a key technological characteristic: its first generation (2004–2015) detected only mechanical clearings, while its second generation (post-2015) began to identify fire disturbances, but with much weaker detection capacity than for mechanical clearing.

I argue that because DETER detects mechanical clearing more effectively than fire-based clearing, it may have favored the fire-based method. To show this, I present a simple theoretical model in which a farmer chooses a deforestation technology based on (i) weather con-

ditions, since fire is only viable when conditions are adequate to burn, and (ii) the perceived punishment probability associated with each technology. Under favorable forest-burning conditions, if farmers perceive a gap between punishment probabilities, land-clearing behavior will strategically shift toward fire-based. In contrast, on-the-ground enforcement presence, triggered by DETER, might create general deterrence by increasing the perceived punishment risk of all clearing technologies, reducing all forms of illegal deforestation. Crucially, perceptions about the punishment probability are not limited to farmers directly affected by enforcement, as observing neighbors' punishments for illegal clearing can shift a farmer's expectations about the relative risk of each method. These two mechanisms produce opposing predictions for fire use: general deterrence would reduce it, while strategic substitution would increase it.

To identify the causal effect of enforcement on deforestation-technology substitution, I use cloud cover as an instrument for enforcement, as proposed in Assunção et al. (2023a). The instrument relies on the fact that the DETER satellite monitoring system cannot see through clouds; therefore, municipalities with higher cloud cover receive fewer immediate alerts, face less on-the-ground enforcement, and receive fewer fines. Although the system eventually detects disturbances and triggers an alert once clouds dissipate, in the Brazilian context, immediate response to ongoing deforestation is crucial for issuing fines. Hence, cloud cover introduces plausible random variation in enforcement exposure across municipalities. Because cloud cover could otherwise be correlated with the weather conditions that make fire a viable clearing method, I control directly for local flammability conditions, so that the remaining variation in cloud cover reflects satellite visibility rather than burning conditions. Comparing municipalities with high versus low cloud cover, and thus low versus high enforcement presence, allows me to isolate the effect of exposure to enforcement on land clearing decisions.

My empirical analysis combines monthly municipal-level data on environmental law enforce-

ment (deforestation fines), cloud cover, and fire ignitions between 2007 and 2018. Across both phases of DETER, I find that a one-unit increase in deforestation fines in a given municipality-month is associated with 5.0 additional fire ignitions in forest-covered areas the following month (both $p < 0.01$). This result indicates that municipalities under stricter enforcement shifted land-clearing behavior toward fire, consistent with monitoring-induced changes in deforestation behavior. The effect is positive and statistically significant for exposure windows of up to six months and fades for longer windows.

Breaking down my results by DETER generation, the effect of enforcement under the first, fire-blind generation is positive and significant (7.65, $p < 0.01$). The change under the second, fire-aware generation is negative, as the substitution hypothesis predicts, but small and statistically insignificant (-0.41, s.e. 1.00). Hence, I find no evidence that the substitution effect weakened under the fire-aware generation. This comparison neither confirms nor rejects the substitution hypothesis, because I demonstrate that DETER's ability to detect fire remained limited after 2015, so the test may lack power to detect a partial narrowing of the detection gap.

My paper's main contribution is the differential effects of conservation policy across deforestation technologies in the Amazon, a key area for conservation in the tropics. This distinction matters because fire can have unique and severe consequences for forest regeneration: unlike mechanical clearing, burned areas risk being locked into a permanent grassland state, preventing forest recovery (Drüke et al., 2023). This is particularly consequential given growing evidence that the Amazon may be approaching a tipping point (Lovejoy and Nobre, 2018), with fire identified as a key compounding disturbance that could trigger an irreversible transition to savanna (Flores et al., 2024). The health externalities are equally substantial. Koplitz et al. (2016) estimates that haze from a single fire season, Indonesia's 2015 forest fires, resulted in 100,300 excess deaths. Taken together, the health and environmental externalities of fire use make understanding what drives individuals to use fire a

question with direct policy relevance.

My work also speaks to the literature on behavioral responses to environmental regulation. Gibson (2019), for example, shows that, in response to the Clean Air Act, firms substitute pollution to other media, i.e., waterways. While most of the literature focuses on the substitution induced by air pollution regulation (Fowlie, 2009; Danae Hernandez-Cortes, 2025; Costa et al., 2025; Zivin et al., 2026), to the best of my knowledge, my work is the first to propose substitution across deforestation technologies due to incomplete monitoring. Understanding agents' responses to regulatory incentives is central to designing environmental regulation, since an efficient policy should account for adaptive responses. However, in many contexts, it is difficult to observe the environmental outcomes of interest needed to assess a potential substitution effect. Remotely sensed data, paired with detailed enforcement information, allow me to observe the effect of regulation in an unexplored context.

This paper also contributes to the literature on monitoring and deterrence of illegal deforestation in the Brazilian Amazon. While previous work has documented how satellite monitoring and environmental regulation reduced deforestation in the mid-2000s (Assunção et al., 2023a,b), new evidence suggests these policies were only partially effective, as agents adapted their behavior to evade detection. For instance, Baragwanath and Shinde (2025) documents how deforestation shifted toward smaller patches to evade satellite detection, and Cammelli et al. (2026) shows that these policies did not deter forest degradation. My work speaks to these findings by explicitly identifying substitution across deforestation technologies, suggesting that the apparent success of conservation policies adopted in the 2000s may be overstated if observed reductions stem, in part, from displacement into less detectable fire degradation. Accurately distinguishing between the two is crucial, as overlooking potential behavioral changes risks overstating enforcement effectiveness and misdirecting policy.

This project also contributes to the literature on the drivers and control of forest fires in the developing world. While Balboni et al. (2021) and Balboni et al. (2023) focus on fire

use in Indonesia, no prior study has examined how environmental enforcement shapes fire use in the Brazilian Amazon, a region of global conservation importance. Although Araujo et al. (2024) is among the few economics papers addressing this context, it focuses on how media attention influences government action in response to a single fire episode in August 2019 and does not address the broader, longer-term effects of enforcement. In parallel, my research complements the natural sciences literature, which predominantly investigates the roles of climate phenomena and weather events in determining fire incidence (Aragão and Shimabukuro, 2010; Libonati et al., 2021; Valente and Laurini, 2023; Flores et al., 2024). By identifying enforcement incentives as an additional driver of fire use, my work introduces a policy-relevant mechanism unexplored by the economics and natural sciences literatures.

The remainder of this paper is organized as follows. Section 2 describes how environmental law enforcement operates in the Brazilian Amazon. Section 3 presents a theoretical model of the trade-off between land clearing technologies. Section 4 describes the data sources, details forest fire patterns, and examines their relationship with land use and land clearing. Section 5 presents the main empirical strategy, Section 6 presents my main results and robustness checks, Section 7 presents additional results, and Section 8 concludes.

2 Background

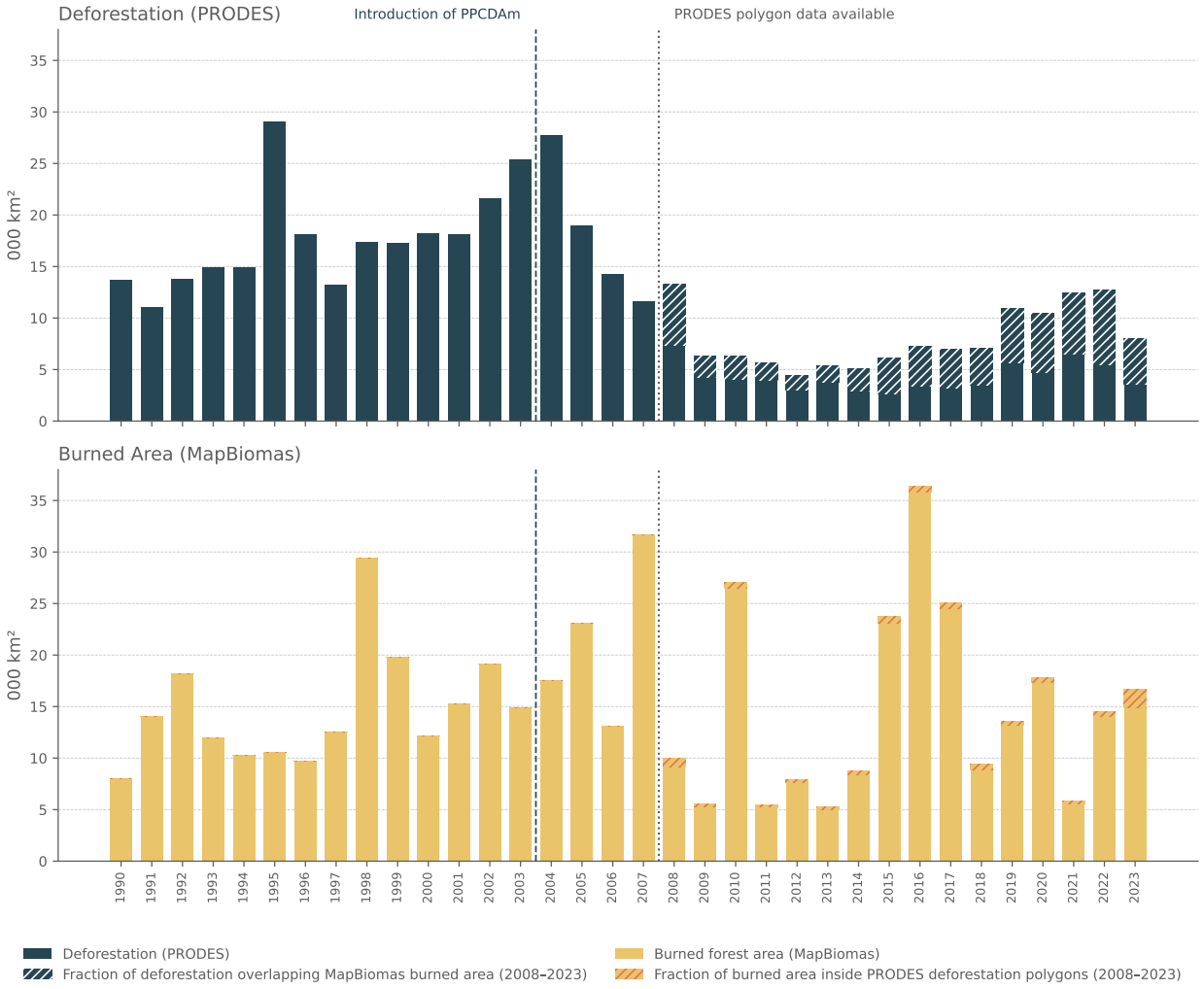
2.1 Deforestation and fires in the Brazilian Amazon

The Brazilian Forest Code requires property owners in the Amazon to maintain 80% of their land as native vegetation, and illegal deforestation, regardless of the technology used, constitutes an environmental infraction subject to administrative and criminal sanctions (Brasil, 2012). IBAMA (Brazilian Institute for the Environment and Renewable Natural Resources) is the federal agency responsible for monitoring and enforcing these rules, with authority analogous to a national environmental police force.

In 2004, Brazil established PPCDAm, the Action Plan for the Prevention and Control of Deforestation in the Legal Amazon, which encompassed multiple anti-deforestation policies and is associated with the sharp reduction in deforestation starting in the mid-2000s (West and Fearnside, 2021). The plan included, for example, blacklisting municipalities with the highest deforestation rates and subjecting them to more intensive environmental monitoring (Assunção and Rocha, 2019), restricting agricultural credit (Assunção et al., 2020), and, most relevant to this paper, launching DETER, a near-real-time, satellite-based deforestation monitoring system. Fundamentally, DETER detects changes in forest cover and triggers alerts to guide on-the-ground enforcement by IBAMA.

Despite PPCDAm being credited with a significant reduction in deforestation after 2004 (West and Fearnside, 2021), the Brazilian Amazon burned, on average, roughly 15.8 thousand square kilometers of forest annually between 2005 and 2023. In most years, this exceeded the area officially recorded as deforested by PRODES, the official Brazilian government statistic for Amazon deforestation. Figure 1 illustrates these trends. While the official deforested area fell sharply after 2004 (top figure), burned forest areas did not decline similarly (bottom figure). Crucially, the bottom figure shows that most burned areas were not included in that year's official deforestation polygons from PRODES, implying that end-of-year deforestation rates alone do not tell the full story of conservation policies' effects.

Figure 1: Annual deforestation and burned forest area in the Brazilian Amazon, 1990–2023

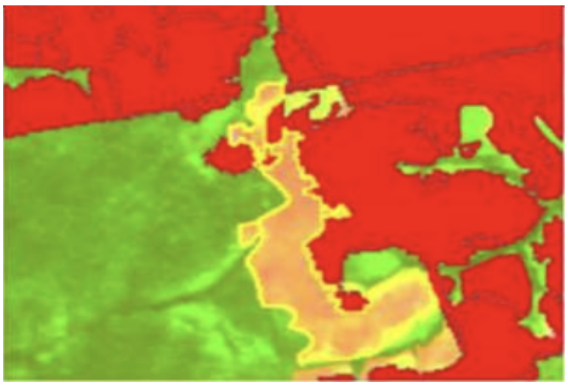


Note: The top panel shows annual deforestation area from PRODES, Brazil's official end-of-the-year statistic. Through 2007, bars show the official annual deforested area in the Legal Amazon. From 2008, when polygon-level data become available (dotted line), bars show the sum of PRODES deforestation polygons, and the hatched portion is the share of that area that overlaps MapBiomas Fogo burned area (within any MapBiomas land-cover class). A polygon area counts as burned if a burn scar is detected there in the calendar year in which its PRODES year (August to July) ends or in the preceding calendar year. The bottom panel shows annual burned forest area in the Amazon biome from MapBiomas Fogo; from 2008, the hatched portion is the share that falls within PRODES deforestation polygons for the same year. In both panels, bar height is the full annual total, so the hatching does not change the height of the bars. The dashed vertical line marks the introduction of the PPCDAm in 2004. Between 2008 and 2023, about 30 to 58% of PRODES deforestation area overlaps burned area, whereas only about 2 to 11% of burned forest area lies inside PRODES polygons. The two shares are not mirror images because the top panel counts burned area in any MapBiomas land-cover class, while the bottom panel counts only fires in forest areas. The differences in land-cover classification methodology between PRODES and MapBiomas explain this difference.

In the Amazon context, fire is predominantly intentional, as lightning-ignited fires are uncommon in the forest because of its humidity. I find that, on average, only about 10% of fire events in the region are preceded by a lightning strike (Figure C.1). While climate events, notably El Niño-driven droughts, amplify fire incidence by reducing forest moisture and facilitating spread (Aragão et al., 2018; Valente and Laurini, 2023), fires have remained closely tied to land degradation over the past two decades, with human behavior the critical determinant (Libonati et al., 2021). Because using fire to clear forests is illegal in Brazil, the intentional nature of fire use makes enforcement incentives a relevant determinant in this setting.

Different deforestation technologies differ in relevant ways from an enforcement perspective. First, satellite images of mechanical clearing and fire-based deforestation differ from the DETER satellite monitoring perspective, as shown in Figure 2. Mechanical deforestation involves the rapid and complete removal of forest cover, leaving exposed soil that contrasts sharply with surrounding vegetation (Figure 2a). This spectral signature is relatively straightforward to detect via DETER (Almeida et al., 2022). Mechanical clearing was the dominant form of deforestation in the Amazon from the 1970s through the early 2000s, so DETER targeted it when it was developed in 2004. Fire-based degradation, by contrast, unfolds over multiple years through successive burning cycles: pasture is seeded into partially cleared forest while trees remain standing; cattle are introduced as the forest canopy thins; the pasture is burned in a second year, further clearing what remains; and by the third year, repeated fires have destroyed the original forest cover (Almeida et al., 2022). The result is a far subtler satellite signature than clear-cutting (Figure 2c), which can easily be confused with natural variation and poses a significant challenge for satellite detection.

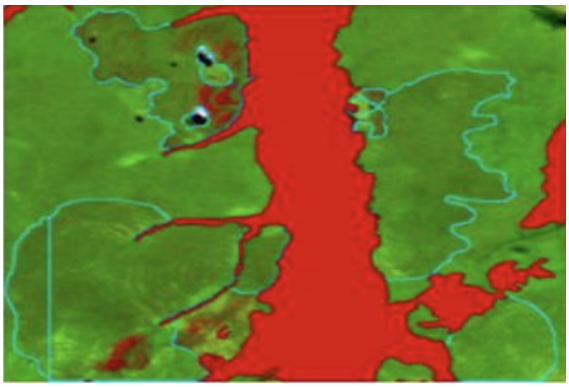
Figure 2: DETER satellite composites and field images



(a) Satellite composite of mechanical clearing — light orange area with yellow outline.



(b) Field image of the corresponding mechanical clearing.



(c) Satellite composite of fire scar — light green outline.



(d) Field image of the corresponding fire scar.

Note: Images from INPE in Valeriano et al. (2016). The red areas represent the PRODES mask used by the DETER system, where areas previously mapped as deforested are excluded.

Another critical asymmetry between the two technologies concerns legal attribution. Whereas mechanical clearing can be unambiguously attributed to human activity, fire has a positive natural probability of occurrence. Brazilian law requires that a causal nexus be established between an individual's action and a specific fire before sanctions can be applied (Brasil, 2012), and satisfying this standard requires extensive investigation (IBAMA, 2010). This legal burden of proof is substantially lower for mechanical clearing, making fire not only

harder to detect but also harder to punish even when detected.

The sanctions for clearing and burning native vegetation in Brazil are set by Decreto 6.514/2008. Between December 2008 and September 2024, the statutory fine for clear-cutting native forest outside a legal reserve without authorization was R$1,000 per hectare (Art. 52), rising to R$5,000 per hectare inside a legal reserve (Art. 51) and R$5,000–50,000 per hectare in permanent preservation areas (Art. 43). Forest clearing carried out with fire was penalized more heavily through a 50% surcharge on the clearing fines (Art. 60). After detecting illegal deforestation, authorities must impose an embargo on the affected area (Art. 16). An embargo is a precautionary administrative measure that bars any economic activity in the specific area where an environmental infraction occurred until the landowner regularizes it with the environmental authority. Violating an embargo carried a fine of R$10,000–1,000,000 (Art. 79). These figures are statutory text and should not be read as the sanctions farmers faced in practice. They are per-hectare amounts that depend on how the enforcement officer classifies the infraction and measures the area, and are suspended pending appeal. They say nothing about the fines actually imposed or collected.

2.2 The DETER Monitoring System

The enforcement of forest conservation rules has been severely constrained by the geographic scale and remoteness of the Amazon region. Because of this monitoring challenge, INPE (Brazil's National Institute for Space Research) launched DETER in 2004 to support PPCDAm. The near-real-time satellite monitoring system detects changes in forest cover and triggers alerts for on-the-ground enforcement by IBAMA.

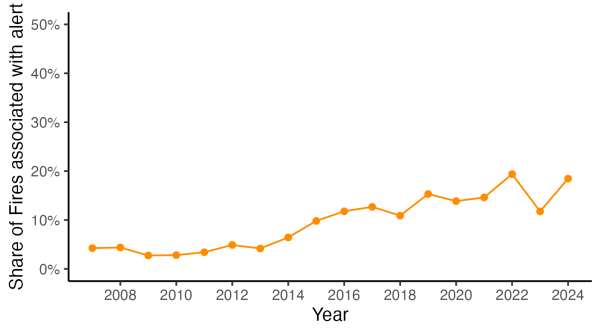
DETER first generation (2004-2015) DETER's trial phase was introduced in 2004, but it became fully operational in 2006. In its initial version, the system relied on near-daily MODIS imagery to detect deforestation events exceeding 25 hectares. The system operated

on an exclusion mask — previously mapped deforestation from PRODES, non-forest areas, and hydrographic features — so that only new alterations to forest cover could generate alerts (Almeida et al., 2022). This first generation had two critical blind spots. First, it missed all clearing events below 25 hectares — in this sub-threshold range, DETER detected only 0.05% of total deforested area (Diniz et al., 2015). Second, and key for this paper, it was nearly blind to fire-based degradation, with burning vegetation generating almost no alerts (Figure 3a). Evidence suggests that at least the first blind spot was strategically explored. Baragwanath and Shinde (2025) show that the average size of deforestation patches declined after DETER's introduction, consistent with land users adjusting their behavior to remain below the detection threshold. Fire, which leaves a qualitatively different spectral signature than mechanical clearing, offered an additional margin of evasion that the system was technically unable to flag.

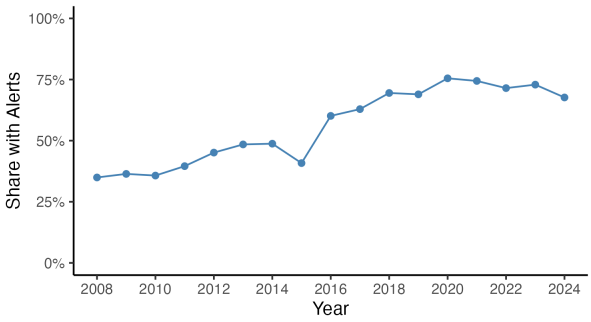
DETER second generation (post-2015) In 2015, INPE introduced DETER-B, which uses higher-resolution imagery (Diniz et al., 2015; Valeriano et al., 2016). DETER-B extended detection to events as small as 3 hectares and, critically, added the capacity to identify fire scars alongside mechanical clearing [1]. Visual image interpretation by trained analysts remains foundational to both generations, reflecting a deliberate institutional choice to minimize false positives (Soler et al., 2021). The reliance on human visual interpretation and the associated learning process helps explain progressive improvement in the system's accuracy (Figure 3b).

[1] With DETER-B, alerts are organized into two groups: deforestation warnings (clear-cut deforestation, deforestation with remaining vegetation, and mining) and degradation warnings (degradation, geometric selective logging, disorganized selective logging, and forest fire scars). Polygons initially mapped as degradation can be reclassified as deforestation in subsequent monitoring cycles, allowing the system to capture progressive forest loss over time.

Figure 3: DETER detection rates for fire versus deforestation, 2007–2024



(a) **Share of fire events with a coincident DETER alert.** For each year, the percentage of MODIS-detected fire events that coincided with a DETER system alert.



(b) **DETER's detection rate of PRODES-confirmed deforestation.** For each year, the percentage of PRODES-confirmed deforestation area that received a corresponding DETER alert.

Note: Deforestation refers to PRODES's official end-of-year deforestation measurement. Panel (a)'s share remains well under 20% throughout the sample, especially reflecting DETER's blind spot for fire-based clearing prior to 2015. Panel (b)'s share rises from roughly 35% (2008) to about 75% by the early 2020s, reflecting improving detection accuracy for mechanical clearing, notably after DETER-B's 2015 upgrade.

In my empirical strategy, I explore two features of the DETER system. First, a constraint affecting both generations: the system can only detect forest disturbances in areas not obscured by clouds. Cloud cover systematically delays the interval between a deforestation event and a satellite alert and, in turn, reduces the likelihood of on-the-ground enforcement (Section 2.3 details it). For this reason, following Assunção et al. (2023a), I use cloud cover as an instrument for enforcement intensity: municipalities with higher cloud cover were less likely to receive timely enforcement, providing exogenous variation in the likelihood of punishment for mechanical clearing. Crucially, under the first generation, cloud cover affected only enforcement of mechanical clearing—fire was already undetectable regardless of cloud conditions —so the instrument isolates variation in clear-cut enforcement alone. Second, I exploit the shift from the first to the second generation, which reduced the detectability gap

between mechanical clearing and fire[2]. Together, these two sources of variation allow me to identify whether satellite detection shaped the relative attractiveness of fire as a land-clearing strategy.

Assunção et al. (2023a) show that DETER was central to the post-2004 decline in PRODES deforestation. Their analysis, however, is limited in two ways. First, it evaluates the system only on the outcome it was designed to reduce, annual PRODES deforestation. Second, the analysis covers 2006-2016, almost exclusively DETER's first generation. As a result, it speaks to whether DETER and PPCDAm achieved their intended goal, but not to how landowners potentially adapted to the system. My paper asks a different question about the unintended consequences of monitoring. Because I use fire ignitions as an outcome and cover both DETER generations, I can evaluate the system's effectiveness at different margins.

2.3 Importance of near-real-time enforcement response to a deforestation event

The severity of sanctions that environmental law enforcement can apply in Brazil depends critically on the timing of the response to a deforestation event (Assunção et al., 2023a). Before DETER, enforcement officers struggled to locate and access new deforestation hotspots in a timely manner, since identifying new clearings relied either on Ibama's ability to accurately anticipate spatial deforestation patterns or on hotline reports. By the time officers reached a deforested area, the window for applying the most binding penalties had typically closed. Even if officers could correctly identify and locate the responsible parties, their ability to impose the costliest penalties ultimately depended on catching offenders on the spot. For example, if law enforcement officers find heavy machinery, such as tractors, on site in a deforestation hotspot, they can impose an immediate and severe financial loss on the

[2]Reduced but did not fully close the gap since (i) as shown in Figure 3a, the detection accuracy of fire events improved post-2015 but is still significantly lower than the accuracy of mechanical clearing; and (ii) *de facto* legal punishment remained relatively more challenging with fire.

offender by seizing and destroying it (Assunção et al., 2013).

The adoption of near-real-time, satellite-based monitoring through DETER changed this dynamic by enabling timely detection. However, the system's inability to detect deforestation through cloud cover lengthens the gap between a deforestation event and an alert. Hence, although new deforestation events will eventually trigger an alert, a longer gap between detection and enforcement makes caught-in-the-act operations less likely.

3 Theoretical framework

The following simple model presents the idea of substitution between deforestation technologies, which informs my main empirical approach. The key feature is that the two technologies differ along two margins: detectability by satellite monitoring, and dependence on environmental conditions. Fire is harder to detect and cheaper to use, but it only works when the forest is dry enough to burn.

Consider a profit-maximizing farmer endowed with a fixed amount of land L [3]. The land is partitioned between:

$$L = l_p + l_f \tag{1}$$

where l_p is the amount of land under agricultural production and l_f is the land kept as forest. Under the Brazilian Forest Code, the farmer must keep 80% of their parcel as native vegetation, such that $\bar{s}_f = 0.8$, where $s_f \equiv l_f/L$. Compliance therefore implies $l_p = l_0 \equiv (1 - \bar{s}_f)\, L = 0.2\, L$. The farmer's problem is to decide whether to comply and keep $s_f = \bar{s}_f$, or to expand production by converting forested land, i.e. to set $s_f < \bar{s}_f$.

[3] The model set-up was heavily inspired by Lee and Barrett (2001).

Production

Given productive land l_p, the farmer chooses the level of a composite input h (fertilizer, labour, machinery, etc.) to maximise the flow of land rent:

$$R^*(l_p) \;=\; \max_{h\geq 0}\Big[\, p\,A\,f(l_p,\,h) \;-\; q\,h\,\Big], \tag{2}$$

where p is the output price, A is total factor productivity, q is the unit cost of the input h, and $f(\cdot)$ is increasing in both arguments ($f_l, f_h > 0$) and concave. Hence $R^{*\prime}(l_p) > 0$, so additional productive land raises the land rent.

For simplicity, let Δl be a marginal conversion, and assuming it is small relative to l_0, I approximate the rent around the compliance level as:

$$R^*(l_0 + x) \;\approx\; R^*(l_0) + \rho\; x, \qquad \rho \equiv R^{*\prime}(l_0) > 0. \tag{3}$$

Deforestation choice

A farmer who chooses to violate the Forest Code must also choose how to clear the additional forested land, $\Delta l \equiv (\bar{s}_f - s_f)L > 0$. Let $t \in \{\textit{fire},\, \textit{cut}\}$ index the available clearing technologies.

The institutional feature discussed in this paper is that the two technologies differ in their detectability by Brazil's DETER satellite monitoring system. That leads to the following assumption:

Assumption 1 (Differential detection risk)**.** *The probability of enforcement conditional on illegal clearing is strictly lower for fire-based clearing than for mechanical cutting:*

$$\theta_{\textit{fire}} \;<\; \theta_{\textit{cut}}, \qquad \theta_t \in [0,1]. \tag{4}$$

This assumption reflects that DETER's first generation detected the spectral signatures of mechanical clearings but could not detect fire-related deforestation. Because DETER detection is the primary trigger for on-the-ground inspections by IBAMA, a lower θ_t implies a lower expected punishment burden. Detection depends on satellite visibility, which is obstructed by cloud cover $c \in [0, 1]$:

$$\theta_t \;=\; \theta_t(c). \tag{5}$$

Under DETER's first generation,

$$\frac{\partial \theta_{fire}}{\partial c} = 0, \qquad \frac{\partial \theta_{cut}}{\partial c} < 0. \tag{6}$$

Costs. Clearing with fire is cheaper than mechanical clearing, which requires equipment and labour. Let k_t denote the cost per hectare of attempting to clear with technology t.

Assumption 2 (Fire is the cheaper deforestation technology)**.** $k_{fire} < k_{cut} < \rho$.

The last inequality guarantees that, absent enforcement, illegal clearing is profitable.

Environmental constraint on fire. The second feature is that fire depends on environmental conditions. Let $w \in [0, 1]$ denote flammability, which captures vegetation dryness. Flammability is exogenous to the farmer and observed before the clearing decision. A fire attempt on Δl hectares only converts a share $\phi(w)$ of the area to agricultural land:

$$\phi(w) \in [0, 1], \qquad \phi'(w) > 0 \text{ for } w > \underline{w}, \qquad \phi(w) = 0 \text{ for } w \leq \underline{w}. \tag{7}$$

When vegetation is too wet to burn ($w \leq \underline{w}$), fire cannot be used to clear land. As conditions dry, a larger share of the intended area burns. Mechanical cutting converts the entire Δl regardless of w. The share $1 - \phi(w)$ of the area that fails to burn yields no agricultural rent,

but the farmer still pays the cost of the attempt.[4]

Sanction. If illegal clearing is detected, the farmer pays a sanction that is linear in the deforested area:

$$F(\Delta l) \;=\; \tau\,\Delta l, \qquad \tau > 0, \tag{8}$$

where τ is the sanction per hectare.[5]

Farmer's Problem

I show the payoffs net of the rent under compliance, $R^*(l_0)$. Using (3), the net gains from each option are:

Compliance

$$G_{comply} = 0. \tag{9}$$

Violation with Fire

With probability θ_{fire} the illegal clearing is detected, triggering the sanction $\tau\Delta l$. The fire-deforestation gain is:

$$G_{fire}(w) \;=\; \Big[\rho\,\phi(w) - k_{fire} - \theta_{fire}\,\tau\Big]\Delta l. \tag{10}$$

Violation with Mechanical Clearing

The sanction per hectare is the same as for fire, but detection is more likely:

$$G_{cut} \;=\; \Big[\rho - k_{cut} - \theta_{cut}\,\tau\Big]\Delta l. \tag{11}$$

[4] In this version I don't include potential private damages from fires escaping into private productive land.

[5] This mirrors the legislation, in which fines are proportional to the hectares of cleared forest. I assume the same per-hectare sanction applies regardless of the clearing technology.

Deforestation technology choice

The farmer chooses fire if $G_{fire}(w) > \max\{G_{cut}, 0\}$. The first condition compares the two technologies. From equations (10) and (11), $G_{fire} > G_{cut}$ if and only if

$$\left(\theta_{cut} - \theta_{fire}\right)\tau \;>\; \rho\left[1 - \phi(w)\right] - \left(k_{cut} - k_{fire}\right). \tag{12}$$

The left-hand side is the expected sanction per hectare saved by using the less detectable technology. The right-hand side is the net cost of fire per hectare: the rent lost on the share of area that does not burn, $\rho[1 - \phi(w)]$, minus the clearing cost saved, $(k_{cut} - k_{fire})$.

The second condition, that fire is better than compliance, requires

$$\rho\,\phi(w) \;>\; k_{fire} + \theta_{fire}\,\tau. \tag{13}$$

When $w \leq \underline{w}$, $\phi(w) = 0$ and condition (13) fails: fire is never chosen when vegetation is too wet to burn. Since the left-hand side is increasing in w, there is a flammability threshold $w^* > \underline{w}$, defined by $\rho\,\phi(w^*) = k_{fire} + \theta_{fire}\,\tau$, above which (13) holds.

Proposition 1 (Determinants of technology choice)**.** *Conditional on flammability $w \geq w^*$, fire becomes relatively more attractive than mechanical cutting when:*

(i) The detection gap $(\theta_{cut} - \theta_{fire})$ is larger;

(ii) The sanction per hectare τ is larger. Stronger punishment raises the comparative advantage of the low-detection technology;

(iii) Flammability w is higher. Drier conditions raise the share of the intended area that burns;

(iv) The cost advantage of fire, $(k_{cut} - k_{fire})$, is larger.

Proposition 1(i) is the main prediction linking the introduction of DETER to fire use. When

monitoring improves its ability to detect clear-cuts but not fire, the gap ($\theta_{cut} - \theta_{fire}$) widens, and fire becomes more attractive. Once DETER can detect fire, θ_{fire} rises and the gap narrows.

Environmental gating of substitution Flammability determines where substitution toward fire is possible. If $w < w^*$, condition (13) fails and fire is not chosen, regardless of the detection gap.

4 Data

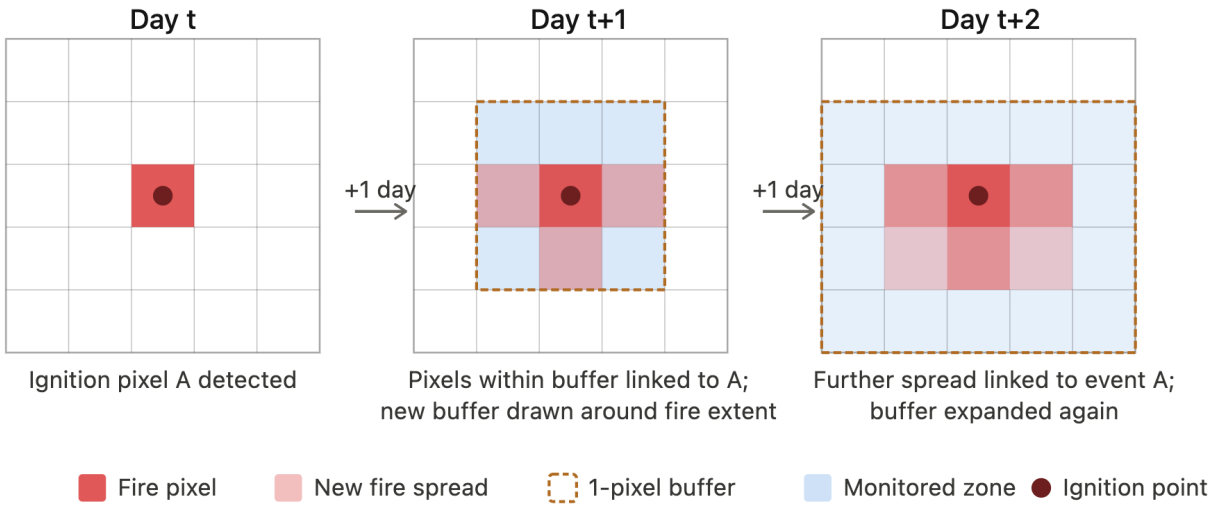
4.1 Fire ignitions

I use the MODIS Terra Thermal Anomalies and Fire product to count fire ignitions. The product presents 1km gridded information on pixels where fire was detected for every 24-hour period. The dataset tracks fire hotspots from 2000 to the present. For this paper, I chose to track the number of fire ignitions instead of directly using MODIS hotspots. The "conversion" of fire ignitions into fire spread depends on climate variables, such as average temperature and soil moisture. Therefore, the number of ignitions better reflects individuals' intentional start of a fire.

To calculate the number of individual fire ignition events in this dataset, I use the procedure proposed by Balboni et al. (2024). The procedure tracks daily MODIS fire observations across time to determine the ignition and spread of each fire event. For each pixel A_t, or a contiguous cluster of pixels with fire observed on day t, a 1-pixel buffer is created around it. All pixels inside this buffer with fire on day $t+1$ are considered to be a continuation of fire A_t. In the presence of cloud cover, the procedure uses the next day's observation. On day $t+1$, a new buffer is created around the new extension of A_t. All pixels with fire within this buffer on day $t+2$ are treated as a continuation of fire A_t. This process is iterated

across each day so I can link each fire-detected pixel to a specific event. Figure 4 presents a visual scheme of this process. For comparison, I also use fire hotspot data from INPE. INPE (Brazil's National Institute for Space Research) uses data from several satellite sensors to track fires in Brazil.

Figure 4: Representation of the fire hotspot-to-ignition-point linking algorithm



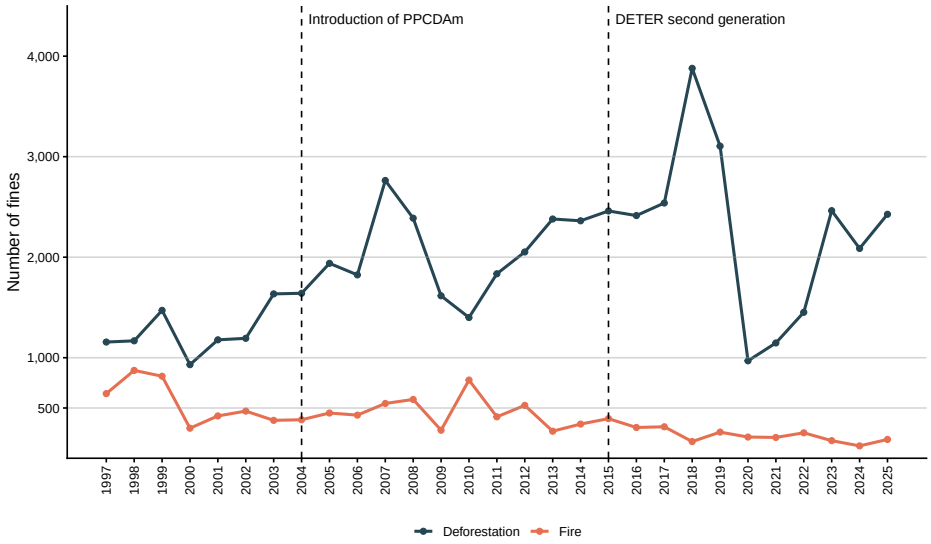
Note: Schematic of the procedure (following Balboni et al., 2024) used to convert daily MODIS fire-pixel detections into discrete ignition events.

Figure C.2 displays the seasonality in fire occurrence in the region. Fire activity follows a pronounced dry-season pattern, peaking sharply in August and September before declining through the wet season (November–April). Figure C.3 shows the distribution of fire ignitions across tenure regimes. Fires are concentrated in properties with full title (in the SIGEF database) or in public lands/non-fully titled areas (others). Conservation units and indigenous lands accounted for a small share of ignitions during the post-DETER period.

4.2 Environmental Law Enforcement data

As in Assunção et al. (2023a), I use fines issued by IBAMA as a proxy for environmental enforcement presence. Unfortunately, there is no data on the exact deployment or tracking of IBAMA's agents' work. Fines are the best proxy because they indicate agents were present at a site and could hold a violator accountable. The fine data includes the infraction type, issuance date, and municipality. Figure 5 shows the number of fines issued annually within the Amazon biome by violation type. I use the detailed crime-type description to create this breakdown. The number of fines associated with fire is systematically lower than that for deforestation. I use this classification to count deforestation-related fines in each municipality each month. This measurement therefore excludes any fines involving fire-related punishment in their description.

Figure 5: Evolution of deforestation-related and fire-related fines



Note: Annual number of fines issued by IBAMA in the municipalities of the Legal Amazon, classified as deforestation- or fire-related, 1997–2025. Data comes from IBAMA's open data on environmental enforcement. The sample does not filter by fine type, cancellation status or availability of coordinates. Duplicate records are dropped. Each fine is assigned to the year of the date the infringer was notified, or the date of the infraction notice when the notification date is missing.

4.3 Cloud cover

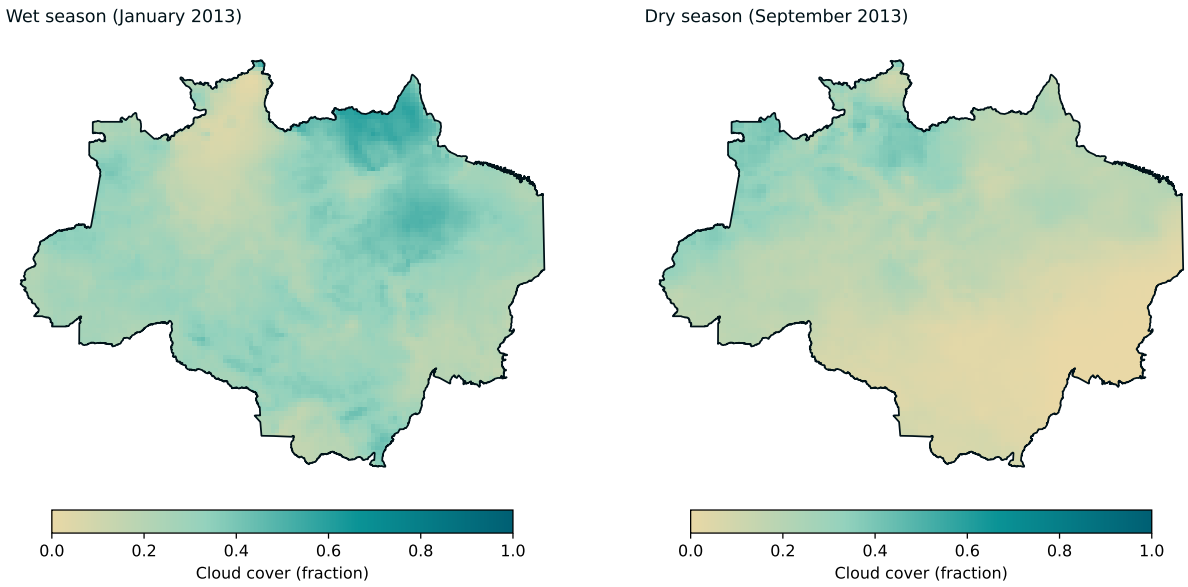
My empirical strategy follows Assunção et al. (2023a) in using cloud cover as an instrument for satellite-based environmental monitoring. When clouds obstruct DETER's satellite view of deforestation scars, the probability of detecting and prosecuting illegal deforestation falls exogenously, generating plausibly random variation in enforcement intensity across municipalities and months.Figure 6 shows the spatial and seasonal variation in cloud cover.

I use two sources of cloud cover data. The primary measure is DETER's operational cloud cover information, which records the fraction of each municipality's area obscured by clouds in each month. However, a key limitation of the DETER cloud measure is in the post-2015 period, when DETER transitioned to its second phase. The second phase uses AWiFS sensors on the Indian ResourceSat-2 and CBERS-4/4A satellites, but cloud cover information is only available for the latter.

For robustness, I also use ERA5 reanalysis data from the European Center for Medium-Range Weather Forecasts. Reanalysis data combine satellite and ground observations with models to fill gaps and provide full coverage. ERA5, among other reanalysis datasets, exhibits the best correlation with MODIS for cloud cover (Wu et al., 2023), while minimizing smoke contamination relative to MODIS. Optical satellite instruments such as DETER can misclassify dense smoke plumes from fires as cloud cover, which would introduce a spurious correlation between the instrument and fire activity—the very outcome of interest for this paper. ERA5 cloud fields are derived from atmospheric reanalysis rather than optical imagery, making them potentially less sensitive to aerosol or smoke loading (as tested in Appendix A.1).

I use DETER cloud cover as my preferred instrument in baseline specifications because it reflects the image blockage the imaging processing team experienced. Table 1 shows that the two measures are highly associated. This close relationship suggests that variation in

Figure 6: Spatial variation in cloud cover across the Brazilian Legal Amazon: wet versus dry season



Note: Panels show ERA5 monthly-mean low cloud cover (fraction of grid-cell area) for January 2013 (wet season) and September 2013 (dry season) — the same product used as the instrument in Section 5. The black outline marks the Brazilian Legal Amazon; darker blue indicates higher cloud fraction. Cloud cover is higher and more spatially uniform in the wet season and drops sharply, especially in the south and east, during the dry/fire season. Source: ERA5 reanalysis, Copernicus Climate Data.

the DETER instrument mostly reflects true meteorological cloud cover rather than smoke-induced misclassification. Figure C.4 shows how the cloud cover variables vary across the seasonal cycle, including during the fire season months (July–October), when smoke contamination is most likely to appear. Finally, one concern about the cloud cover data is whether fire smoke affects it. In Appendix A.1, I show evidence that smoke contamination does not significantly affect the cloud cover measurements used.

Table 1: Association between cloud cover variables

	DETER cloud cover		
	(1)	(2)	(3)
ERA5 low cloud	0.9909***		
	(0.0296)		
ERA5 high cloud		0.3490***	
		(0.0179)	
ERA5 total cloud			0.3946***
			(0.0189)
R^2	0.64215	0.62982	0.63042
Observations	159,785	159,785	159,785
Municipality fixed effects	✓	✓	✓
Year-month fixed effects	✓	✓	✓

Note: OLS regressions of DETER's operational cloud cover measure (share of a municipality's area obscured by clouds in a given month) on three ERA5 reanalysis cloud cover measures, estimated separately: low-level cloud cover (col. 1), high-level cloud cover (col. 2), and total cloud cover (col. 3). Municipality and year-month fixed effects included; standard errors clustered by municipality in parentheses.

4.4 Other sources

Land-use data I use the MapBiomas Brazil land-cover product to track pre-fire land cover information. This allows me to distinguish fires in agricultural versus forest areas.

Weather controls My rainfall data comes from CHIRPS. All additional weather controls come from the ERA5 reanalysis product. Evaporation and soil moisture (0–100 cm) come from ERA5-Land. Table B.2 summarizes the main variables used in the empirical analysis.

5 Empirical framework

5.1 Deforestation technology substitution effect

Understanding how environmental law enforcement affects land-clearing behavior is challenging because of the endogeneity between monitoring efforts and illegal land-clearing activities. Enforcement should be present in areas under deforestation pressure, but it may also deter illegal behavior, so closely monitored areas might experience less illegal deforestation.

To overcome this reverse-causality problem, I propose using the instrument for environmental law enforcement introduced in Assunção et al. (2023a). The instrument leverages the fact that DETER, the near-real-time deforestation alert system, is the primary resource for triggering on-the-ground enforcement in the Brazilian Amazon. Clouds limit the system's capacity to detect clearings. As a result, cloud coverage acts as a plausibly random shifter of enforcement presence. I use this to examine how farmers respond to enforcement exposure, measured by fines issued in a given location and instrumented by cloud cover.

For my estimates, I construct a municipality-month panel from 2007[6] to 2018[7], covering both of DETER's monitoring phases. I estimate the following first stage:

$$E_{i,t-1} = \alpha_i + \omega_t + \sum_{k=1}^{3} \pi_k \, \text{Clouds}_{i,t-k} + \gamma \mathbf{X_{it}} + \epsilon_{it} \tag{14}$$

where $E_{i,t-1}$ is the number of deforestation-related fines in municipality i in month-year $t-1$, and $\text{Clouds}_{i,t-k}$ is the average cloud cover in municipality i in month-year $t-k$, measured with either DETER or ERA5 cloud cover data. I use three lags of cloud cover because the enforcement response to a deforestation event is not immediate. IBAMA must plan and send

[6]DETER was introduced in 2004 but became fully operational only in 2006. Cloud cover data is only consistently available from March 2007.

[7]I choose to limit this analysis to the pre-Bolsonaro period (2019-2022), where DETER's use by IBAMA was severely reduced due to budget cuts.

a field team once the DETER notification is received, and a fine is registered only once the infringer is notified, so the date in the fines data can come well after the infraction. α_i is a municipality fixed effect and ω_t is a month-year fixed effect.

$\mathbf{X_{it}}$ is the set of monthly municipal-level weather controls. My baseline specification includes: rainfall, mean temperature, relative humidity, evaporation, and soil moisture in the top meter (0–100 cm). Evaporation and soil moisture, which come from ERA5-Land. They capture two physical channels through which cloud cover could affect fire independently of enforcement. Because clouds block incoming sunlight, cloudier months have lower evaporation and slower soil drying. Evaporation measures how fast the atmosphere is drying surface vegetation, while soil moisture measures stored dryness, which builds up over several weeks and is only partly reflected in a single month of rainfall. Controlling for both ensures the cloud cover instrument cannot affect fire through these channels, supporting the exclusion restriction in Section 5.2. The second stage is given by:

$$F_{it} = \psi_i + \lambda_t + \beta \hat{E}_{it-1} + \delta \mathbf{X_{it}} + \epsilon_{it} \tag{15}$$

where F_{it} is the number of ignitions in a given municipality. The coefficient β will be positive if enforcement, instrumented in the first stage, increases fire ignitions, indicating substitution. $\hat{E}_{it-1}$ is the number of deforestation-related fines in municipality i in month-year t instrumented by cloud coverage in the first stage. ψ_i is a municipality fixed effect, λ_t is a month-year fixed effect, and $\mathbf{X_{it}}$ is the same set of municipal-level weather controls as in my first stage.

5.2 Identifying assumptions

Exclusion restriction I argue that the exclusion restriction — that cloud cover affects fire incidence only through its effect on enforcement — holds after controlling for rain-

fall, temperature, relative humidity, evaporation, and soil moisture. These weather controls absorb five direct channels through which cloud cover could affect fire use independently of enforcement: a "wet fuel" channel (rainfall), a "moisture content" channel (humidity), a "flammability" channel (temperature), an "atmospheric drying" channel (evaporation), and a "stored dryness" channel (soil moisture). The last two address a radiation channel: cloud cover blocks incoming sunlight, which slows evaporation and soil drying and therefore changes how readily vegetation burns even for the same rainfall. Soil moisture in the top meter (0–100 cm) also summarizes the accumulated effect of weather over previous weeks, which a single month of rainfall does not capture.

A further concern is that cloud cover is correlated with the dry season in the Amazon, which is precisely when fire incidence peaks. My month-year fixed effects absorb aggregate seasonal patterns common across municipalities, while municipality fixed effects absorb time-invariant local differences in fire and cloud seasonality. Together, these controls isolate within-municipality variation in cloud cover that is orthogonal to predictable seasonal fire patterns.

One remaining potential channel is that cloud cover could directly affect farmers' propensity to use fire by signaling reduced satellite visibility, inducing more fire use independently of whether enforcement actually decreases. I argue this channel is unlikely to be quantitatively important, as farmers are unlikely to observe cloud cover at the precision needed to time fire-setting. Conditional on weather controls and fixed effects, cloud cover is therefore plausibly a pure measure of satellite opacity, affecting fire use only through its effect on DETER's detection capacity and the enforcement it triggers.

Instrument relevance Table 2 shows variations in my first-stage estimates across different cloud-cover measurement leads and lags. Contemporaneous and lagged DETER cloud cover have a statistically significant negative correlation with the number of fines within a

municipality. Crucially, lead cloud cover has no statistically significant effect on fines, providing a clean falsification: if cloud cover were proxying for some omitted variable correlated with enforcement, we would expect future cloud cover to predict current fines, which it does not. Figure C.5 displays the association between cloud cover leads and lags on the number of deforestation-related fines issued by IBAMA. First-stage F-statistics are reported in all second-stage tables and exceed the conventional threshold ($F > 10$) in the main specifications (Section 6).

Table 2: First-stage estimates: cloud cover leads and lags on enforcement fines

Dependent Variable:	Deforestation fines		
Model:	(1)	(2)	(3)
Variables			
DETER cloud coverage $_{t}$	-0.059**		-0.065**
	(0.027)		(0.027)
DETER cloud coverage $_{t-1}$		-0.062***	
		(0.023)	
DETER cloud coverage $_{t-2}$		-0.101***	
		(0.023)	
DETER cloud coverage $_{t+1}$			0.024
			(0.021)
Fixed-effects			
Municipality FE	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes
Fit statistics			
R^2	0.156	0.159	0.157
Observations	104,326	98,886	100,432
F-test (1st stage)	6.188	14.145	3.368
Climate controls	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes

Clustered (Municipality) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: OLS regressions of the number of deforestation-related fines issued in a municipality-month on cloud cover at different leads and lags: DETER cloud coverage (cols. 1–3) and ERA5 low/total cloud cover (cols. 4–7). Municipality and year-month fixed effects included; standard errors clustered by municipality in parentheses. Contemporaneous and one-month-lagged cloud cover are negatively and significantly associated with fines under both measures. Lead cloud cover (col. 3, t+1) and ERA5 total cloud cover (col. 7) are not significant — the lead result serves as a falsification check on the exclusion restriction.

Smoke contamination A potential threat to instrument validity is that smoke from fires could contaminate my cloud cover measurements, introducing a mechanical correlation between the instrument and the outcome. I address this concern directly in Appendix A.1. I

show that the association between fire ignitions and cloud cover is negative and very small, and the within-municipality R^2 of fire ignitions on cloud cover is close to zero, ruling out meaningful smoke-driven contamination of the DETER and ERA5 cloud cover measurements.

6 Main results

Deforestation-technology substitution effects I find that exposure to environmental law enforcement, measured by deforestation-related fines, increases fire ignitions in the following month. Table 3 reports the 2SLS estimates of Equation (15) for 2007–2018. Column (1) instruments deforestation fines with DETER cloud cover, column (2) with ERA5 cloud cover, and column (3) reports the OLS estimate for comparison. One additional deforestation fine in a municipality-month raises the number of fire ignitions in the following month by 5.0 with DETER cloud cover and by 7.0 with ERA5 cloud cover, both significant at the 1% level. These magnitudes are large relative to the mean of 0.7–0.8 ignitions per municipality-month (Table 8). They should be read against the fact that fines are rare: enforcement is zero in most municipality-months, so a one-fine change is a large shift in enforcement exposure, and the IV identifies the effect for municipality-months in which cloud cover changes whether enforcement occurs.

The OLS estimate is 0.076 (column 3), roughly 65 to 90 times smaller than the IV estimates. This gap is expected for at least two reasons. First, fines are an imperfect proxy for enforcement presence, so measurement error attenuates the OLS coefficient. Second, enforcement is endogenous to deforestation pressure, and the direction of that bias is not signed a priori. The IV estimates are also local to the compliers whose enforcement exposure responds to cloud cover, and need not equal the average effect across all municipality-months.

The instruments are relevant, although not equally so. The first-stage F-statistic is 15.1

for ERA5 cloud cover and 10.5 for DETER cloud cover (Table 2 shows the underlying first stages), and Table B.3 reports the first-stage estimates for the main specification. The DETER first stage is therefore close to the conventional threshold, and I place more weight on the ERA5 specification, which is also my preferred one. The weaker DETER first stage is consistent with measurement noise in DETER cloud cover after 2015. Sargan statistics of 1.05 and 0.55 do not reject the overidentifying restrictions in either column.

Table 3: IV estimates of enforcement effects on fire ignitions, 2007–2018

Dependent Variable:	Fire ignitions (count)		
	IV		OLS
Model:	(1)	(2)	(3)
Variables			
Enforcement (deforestation fines$_{t-1}$)	5.018***	6.967***	0.0755***
	(1.118)	(1.558)	(0.0273)
Fixed-effects			
Municipality FE	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes
Fit statistics			
F-test (1st stage), Enforcement (deforestation fines$_{t-1}$)	10.479	15.124	
Sargan	1.0518	0.54872	
Observations	96,538	106,674	108,220
Instrument	DETER cloud$_{t-1}$	ERA5 cloud$_{t-1}$	—
Climate controls	Yes	Yes	Yes

Clustered (Municipality) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: Two-stage least squares (cols. 1–2) and OLS (col. 3) estimates of the effect of deforestation fines issued in month $t-1$ on the number of fire ignitions in month t in a municipality. Fines are instrumented with lagged DETER cloud cover (col. 1) or lagged ERA5 cloud cover (col. 2); see Equations (14) and (15). Municipality and year-month fixed effects and weather controls are included. Standard errors clustered by municipality in parentheses. The Sargan statistic tests the overidentifying restrictions.

Fire relative to mechanical clearing Higher fire use only indicates substitution if it does not come with a matching rise in mechanical clearing. To test this, I build a measure of the relative use of the two technologies. For each municipality-month, F_{it} is the area of forest lost to fire (km^2) and C_{it} is the area of forest mechanically cleared (km^2), both measured from monthly 30-meter Landsat composites. A forest pixel counts as mechanically cleared

in month t if it was forest in the previous year, shows a bare-soil signature in t but not in $t-1$, and shows no burn signature. It counts as lost to fire if the same transition comes with a burn signature, measured by the change in the Normalized Burn Ratio. I then estimate

$$\ln\left(\frac{F_{it}}{C_{it}}\right) = \psi_i + \lambda_t + \beta \hat{E}_{i,t-1} + \delta \mathbf{X_{it}} + \epsilon_{it}, \tag{16}$$

where $\hat{E}_{i,t-1}$ is instrumented with cloud cover as in Equation (14) and the remaining terms are as in Equation (15). The coefficient β is positive if enforcement shifts the mix of clearing technologies toward fire. Because the log ratio is undefined when either area is zero, this specification uses only municipality-months in which both fire and mechanical clearing are detected, so its sample is smaller than in the main table. Table 4 repeats the IV specification for the log ratio (column 1), the log of the area lost to fire (column 2), and the log of the mechanically cleared area (column 3). Enforcement raises the fire-to-clearing ratio (3.03, $p < 0.01$) and log fire (3.24, $p < 0.01$), while the effect on mechanical clearing is small and statistically indistinguishable from zero (0.21, s.e. 0.46). The ratio therefore moves because fire rises, not because clearing falls, and the clearing estimate is too imprecise to rule out a modest decline. The result corroborates a shift toward fire rather than a one-for-one displacement of mechanical clearing.

Table 4: IV estimates of enforcement effects on fire and clearing outcomes

Dependent Variables:	log(ratio)	log(Fire)	log(clearing)	log(ratio)
Model		IV		OLS
	(1)	(2)	(3)	(4)
Variables				
Enforcement (deforestation fines$_{t-1}$)	3.027***	3.239***	0.212	0.001
	(0.886)	(0.874)	(0.464)	(0.005)
Fixed-effects				
Municipality FE	Yes	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes	Yes
Fit statistics				
F-test (1st stage), Enforcement (deforestation fines$_{t-1}$)	14.090	14.090	14.090	
R^2	0.631	0.609	0.689	0.630
Observations	91,537	91,537	91,537	92,245
Climate controls	Yes	Yes	Yes	Yes

Clustered (Municipality) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: IV estimates (cols. 1–3) and OLS estimate (col. 4) of the effect of deforestation fines in $t-1$ on the log ratio of the area lost to fire to the mechanically cleared area (cols. 1, 4), the log area lost to fire (col. 2) and the log mechanically cleared area (col. 3). Municipality and year-month fixed effects and weather controls included; standard errors clustered by municipality.

6.1 Placebo checks

Variations in fine type

If cloud cover affects fire only through satellite-triggered enforcement, it should predict the fines that DETER alerts generate and should not predict fines unrelated to satellite detection. Table 5 regresses three fine categories on contemporaneous and lagged cloud cover. Cloud cover has a negative and significant effect on deforestation fines at one and two lags. It has the same effect on fire-use fines. This is unsurprising given that fire-use fines usually cite illegal clearing, so the alerts that detect clearing also lead to them.

The "Other fines" category mostly concerns fauna-related infractions, which a satellite cannot observe. The results suggest cloud cover does not predict them: the first stage is weak (F = 1.0), and the coefficients are insignificant, except for a marginally significant one-month lag

(-0.071, $p < 0.1$). The contrast between the two groups supports the interpretation that cloud cover affects enforcement through DETER's detection.

Cloud cover also reduces fire-use fines, which opens a second channel from the instrument to fire. This channel pushes in the opposite direction from the effect I estimate: fewer fire-use fines should, if anything, raise fire use, which makes cloud cover look less negatively related to fire. It therefore biases my estimates toward zero, and my estimates can be read as conservative.

Table 5: Placebo exercise: by type of fine

Dependent Variables:	Deforestation fines	Fire-use fines	Other fines
Model:	(1)	(2)	(3)
Variables			
Deter cloud$_t$	-0.031	-0.005	0.060
	(0.026)	(0.008)	(0.044)
Deter cloud$_{t-1}$	-0.056**	-0.026***	-0.071*
	(0.022)	(0.008)	(0.039)
Deter cloud$_{t-2}$	-0.104***	-0.041***	0.012
	(0.024)	(0.008)	(0.043)
Fixed-effects			
Municipality FE	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes
Fit statistics			
F-test (1st stage)	10.117	22.471	1.015
R^2	0.160	0.089	0.161
Observations	96,538	96,538	96,538
Climate controls	Yes	Yes	Yes

Clustered (Municipality) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: Dependent variables are the number of deforestation-related fines (col. 1), fire-use fines (col. 2), and all other fines (col. 3) in a municipality-month. DETER cloud cover at t, $t-1$ and $t-2$ as regressors. Municipality and year-month fixed effects and weather controls included; standard errors clustered by municipality.

Cloud cover and fines outside DETER's operation

I also check whether cloud cover predicts fines in periods when DETER was not technically or practically operating as a tool to trigger enforcement. Table 6 reports the first-stage relationship for three samples. The 2019–2022 column covers the Bolsonaro administration, when IBAMA enforcement collapsed for political reasons. Cloud cover does not predict fines there (coefficients of -0.15 and -0.23, both insignificant), which is what one would expect if enforcement was no longer responding to DETER alerts. In the 2010–2015 period, a phase of strong environmental law enforcement, the two-month lag is negative and marginally significant (-0.29, $p < 0.1$) and the first-stage F-statistic is 14.2.

In the pre-2006 period, before DETER's operation, the F-statistic is only 2.9. This is consistent with the fact that clouds shouldn't be affecting enforcement, as it was mostly driven by field visits and anonymous reports.

Table 6: Placebo exercise: periods outside DETER's operation

Dependent Variable:	Deforestation fines		
	Pre-2006	Bolsonaro (2019-2022)	2010-2015
Model:	(1)	(2)	(3)
Variables			
ERA5 cloud $_{t}$	0.002	-0.240	-0.560***
	(0.021)	(0.150)	(0.183)
ERA5 cloud $_{t-1}$	-0.041***	-0.154	-0.223
	(0.015)	(0.281)	(0.137)
ERA5 cloud $_{t-2}$	-0.023	-0.227	-0.290*
	(0.030)	(0.300)	(0.151)
Fixed-effects			
Municipality FE	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes
Fit statistics			
F-test (1st stage)	2.884	3.815	14.165
Observations	18,552	37,104	55,656
R^2	0.090	0.186	0.268
Climate controls	Yes	Yes	Yes

Clustered (Municipality) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: Dependent variable is the number of deforestation-related fines in a municipality-month. Regressors are ERA5 cloud cover at $t-1$ and $t-2$. Column (1) uses the pre-2006 period, column (2) the Bolsonaro administration (2019–2022), and column (3) 2010–2015. Municipality and year-month fixed effects and weather controls included; standard errors clustered by municipality.

6.2 Robustness

Alternative fire data Table 7 replicates the main specification with INPE's fire hotspot counts and with fire spread area as dependent variables. The estimates remain positive and significant at the 1% level under both instruments. For INPE hotspots the coefficients are 36.6 (DETER) and 36.8 (ERA5), against an OLS estimate of 0.63. INPE counts hotspots

rather than ignition events, so these magnitudes are not comparable with those in Table 3. For fire spread area (MODIS Data), one additional fine raises forest burned area by 5.9 km^2 (DETER) and 8.3 km^2 (ERA5). The overidentification test rejects in the INPE specification with ERA5 cloud cover (Sargan = 27.5, column 2), so that estimate should be read with caution; it does not reject in the other IV columns.

Table 7: Alternative fire outcomes: INPE hotspots and fire spread area

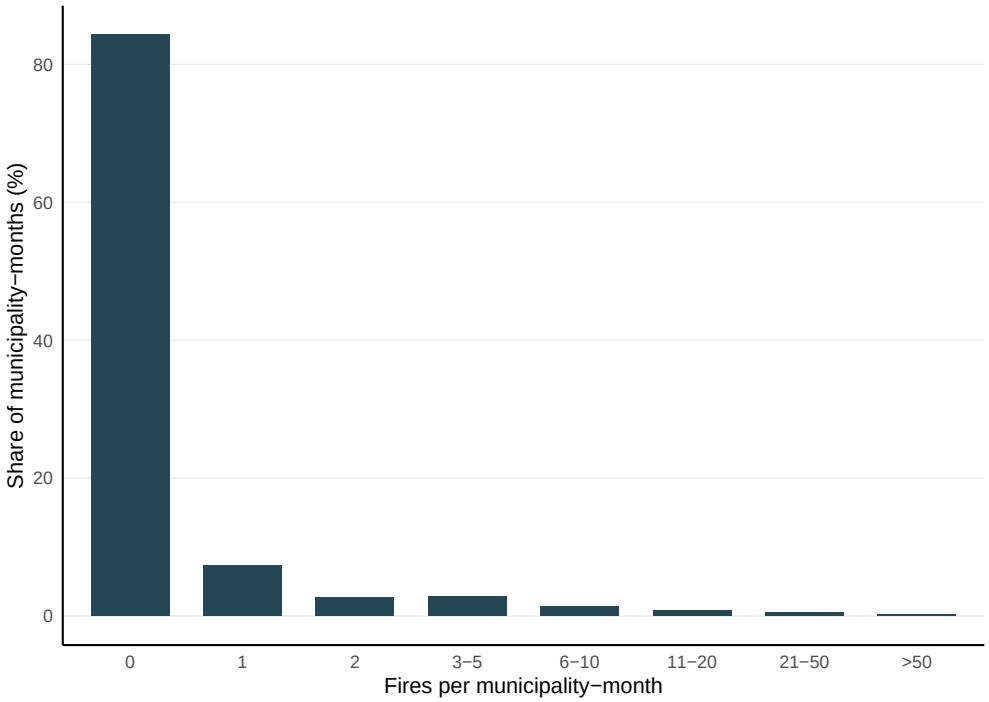
Dependent Variables:	Fire ignitions (INPE count)			Fire spread area (km^2)		
	IV		OLS	IV		OLS
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Enforcement$_{t-1}$)	36.59***	36.79***	0.6269***	5.867***	8.272***	0.0867***
	(7.861)	(9.580)	(0.2413)	(1.325)	(2.008)	(0.0307)
Fixed-effects						
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
F-test (1st stage), Enforcement$_{t-1}$)	10.479	12.849		10.479	12.849	
Sargan	1.6797	27.458		0.94998	0.88378	
Observations	96,538	106,674	108,220	96,538	106,674	108,220
Instruments	DETER cloud	ERA5 low cloud	—	DETER cloud	ERA5 low cloud	—
Climate controls	Yes	Yes	Yes	Yes	Yes	Yes

Clustered (Municipality) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: IV (cols. 1–2, 4–5) and OLS (cols. 3, 6) estimates of the effect of deforestation fines in $t-1$ on INPE fire hotspot counts (cols. 1–3) and fire spread area in km^2 (cols. 4–6). Columns (1) and (4) instrument with DETER cloud cover, columns (2) and (5) with ERA5 low cloud cover. Municipality and year-month fixed effects and weather controls included; standard errors clustered by municipality. Because INPE counts hotspots rather than discrete ignition events, magnitudes are not directly comparable with Table 3.

Robustness to zeros and skewness in the outcome Fire counts are zero in 84% of municipality-months and are highly concentrated (Figure 7). Municipality-months with 11 or more fires make up 1.4% of the sample but 54% of all fires, and the top 1% of observations account for 47%. This raises two concerns about the linear IV estimates in Table 3: a few extreme observations could drive the coefficient, and a linear model may fit a mostly-zero outcome poorly. Table 8 addresses both.

Figure 7: Distribution of fire ignitions per municipality-month



Note: Share of municipality-months by number of fire ignitions. Sample: 106,674 municipality-months, 2007–2018.

Table 8: Robustness to zeros and skewness of the fire outcome

Dependent variable:	Fire ignitions	Fire ignitions (winsorized)	Fire ignitions (winsorized)	Any fire (0/1)
Specification:	Baseline	Winsorized outcome	Winsorized outcome and fines	Linear probability
Model:	(1)	(2)	(3)	(4)
Panel A: IV, DETER cloud cover				
Enforcement (deforestation fines$_{t-1}$)	4.006***	2.177***	4.432***	0.328***
	(0.898)	(0.496)	(0.806)	(0.079)
First-stage F	10.12	10.12	28.95	10.12
Observations	96,538	96,538	96,538	96,538
Mean of dependent variable	0.786	0.564	0.564	0.166
Panel B: IV, ERA5 low cloud				
Enforcement (deforestation fines$_{t-1}$)	3.632***	2.662***	7.114***	0.264***
	(1.113)	(0.776)	(1.701)	(0.089)
First-stage F	13.15	13.15	20.28	13.15
Observations	106,674	106,674	106,674	106,674
Mean of dependent variable	0.732	0.529	0.529	0.157
Panel C: No instrument (OLS)				
Enforcement (deforestation fines$_{t-1}$)	0.075***	0.025***	0.090***	0.002***
	(0.027)	(0.008)	(0.022)	(0.001)
Observations	108,220	108,220	108,220	108,220
Mean of dependent variable	0.725	0.524	0.524	0.156
Municipality FE	Yes	Yes	Yes	Yes
Year-Month FE	Yes	Yes	Yes	Yes
Climate controls	Yes	Yes	Yes	Yes

Clustered (Municipality) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

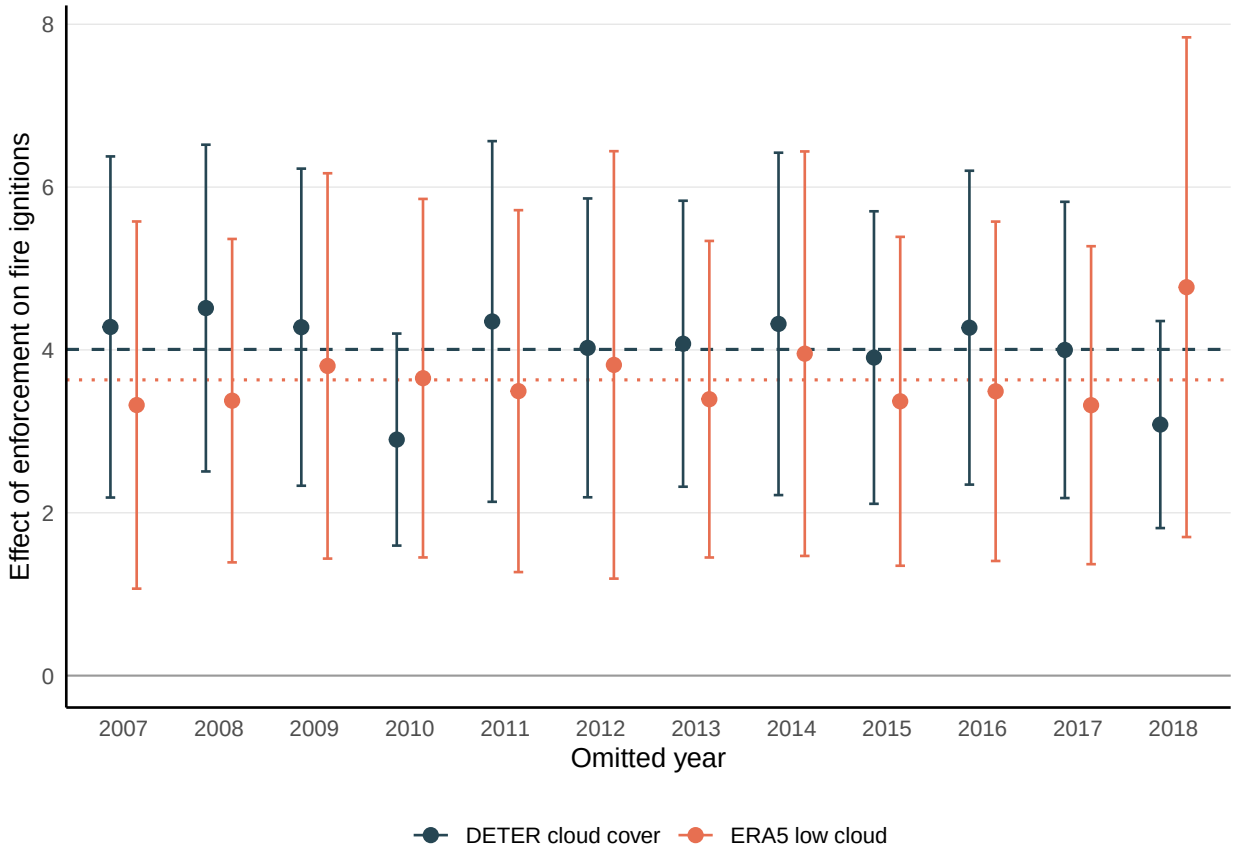
Note: Dependent variable is fire ignitions (count), except in column (4), an indicator for any fire. Columns (2) and (3) winsorize fire counts at the 99th percentile (14 fires); column (3) also winsorizes deforestation fines at the 99th percentile (5 fines). Panels A and B instrument fines in $t-1$ with three lags of the indicated cloud measure; Panel C reports OLS. The sample is 2007–2018. Standard errors clustered by municipality.

Winsorizing fire counts at the 99th percentile (14 fires) lowers the coefficient from 4.01 to 2.18 under DETER and from 3.63 to 2.66 under ERA5 low cloud cover, so part of the baseline magnitude reflects the right tail of the distribution. The effect nevertheless remains positive and significant at the 1% level under both instruments (column 2). Also capping enforcement at its 99th percentile (5 fines) yields larger per-fine coefficients (4.43 and 7.11, column 3), as expected when the treatment is compressed. These are not comparable in units with column (1), but they come with stronger first stages (F = 29.0 and 20.3).

Column (4) examines the extensive margin. One additional fine raises the probability that a municipality-month records any fire by 0.33 under DETER and 0.26 under ERA5 low cloud, against sample means of 0.17 and 0.16. Because enforcement is rarely non-zero, these linear probability coefficients should be read as local average effects rather than as the effect of a single fine for every municipality.

Figure 8 re-estimates the baseline specification omitting one year at a time. The coefficient ranges from 2.9 to 4.5 under DETER and from 3.3 to 4.8 under ERA5 low cloud, and remains positive and statistically significant at the 5% level in every case, so no single year drives the result.

Figure 8: Leave-one-year-out IV estimates



Note: Each point re-estimates the baseline IV omitting the indicated year; bars are 95% confidence intervals clustered by municipality. Horizontal lines show the full-sample estimates (DETER: dashed; ERA5 low cloud: dotted).

7 Additional results and robustness checks

7.1 Effect across the different DETER generations

DETER's second generation, introduced in 2015, extended detection to fire-based clearing. If farmers substitute toward fire because fire falls outside the monitoring system's reach,

the positive effect of enforcement on fire use should shrink once fire becomes detectable. I test this by allowing the effect of enforcement to differ before and after August 2015, the operational rollout of DETER-B's fire detection capability. Let D_t equal one from August 2015 onward and zero otherwise. I instrument both enforcement and its interaction with D_t, using three lags of ERA5 low cloud cover and their interactions with D_t. The first stage is

$$E_{i,t-1} = \alpha_i + \omega_t + \sum_{k=1}^{3} \left[\pi_k \text{Clouds}_{i,t-k} + \phi_k \left(\text{Clouds}_{i,t-k} \times D_t\right)\right] + \gamma \mathbf{X_{it}} + \epsilon_{it} \tag{17}$$

with an analogous first stage for $E_{i,t-1} \times D_t$, and the second stage is:

$$F_{it} = \psi_i + \lambda_t + \beta_1 \hat{E}_{it-1} + \beta_2 \left(\hat{E}_{it-1} \times D_t\right) + \delta \mathbf{X_{it}} + \epsilon_{it}, \tag{18}$$

where all other variables are as defined in Equations (14) and (15). The coefficient β_1 is the effect of enforcement on fire ignitions under DETER-1, when fire was outside the system's detection capability. The coefficient β_2 is the change in that effect once fire becomes detectable, and the substitution hypothesis predicts $\beta_2 < 0$.

Table 9 reports the results, using deforestation fines, the enforcement measure used throughout the paper, in column (1) and all fines in column (2). In column (1), the effect of enforcement under DETER-1 is positive and significant ($\hat{\beta}_1 = 7.65$, $p < 0.01$). The interaction term is negative, as the substitution hypothesis predicts, but small and statistically insignificant ($\hat{\beta}_2 = -0.41$, s.e. 1.00), with a 95% confidence interval from about -2.36 to 1.55. The implied effect after August 2015 is therefore about 7.2. I do not find evidence that the effect attenuated after fire detection was introduced. Column (2) reaches the same conclusion with all fines: the effect under DETER-1 is positive and significant ($\hat{\beta}_1 = 2.35$, $p < 0.01$) and the change after 2015 is statistically insignificant ($\hat{\beta}_2 = 0.48$, s.e. 0.68). The first-stage F-statistic for enforcement in column (1) is 7.7, below the conventional threshold of 10, so I read that column with caution. The first-stage F-statistics for the interaction terms are 19.9 in both

columns, and the F-statistic for all fines in column (2) is 15.7.

This result neither confirms nor rejects the technological-gap mechanism. DETER-B's ability to detect fire was still limited after 2015 (Figure 3a), so the gap in detection probabilities between clearing methods narrowed but did not close, and the test may have limited power to detect a partial narrowing. The estimate also cannot separate the introduction of fire detection from other changes after 2015. The effect of enforcement on fire remains positive in both phases, and the evidence for substitution therefore rests on the main estimates above rather than on the phase comparison.

Table 9: Effect heterogeneity across DETER generations

Dependent Variable:	Fire ignitions	
Model:	(1)	(2)
Variables		
Deforestation fines	7.649***	
	(2.261)	
All Fines fines × Post-2015	-0.4074	0.4776
	(0.9969)	(0.6810)
All fines		2.345***
		(0.6169)
Fixed-effects		
Municipality FE	Yes	Yes
Year-Month FE	Yes	Yes
Fit statistics		
F-test (1st stage), Deforestation fines	7.6982	
F-test (1st stage), All Fines fines × Post-2015	19.935	19.935
F-test (1st stage), All fines		15.697
R^2	0.18145	0.18101
Observations	106,674	106,674
Climate controls	Yes	Yes

Clustered (Municipality) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: IV estimates of the effect of enforcement in $t-1$ on fire ignitions, allowing the effect to differ between DETER's first generation (through July 2015) and second generation (from August 2015), per Equation (18). Enforcement is measured with deforestation fines in column (1) and all fines in column (2). The first row is the effect before August 2015; the second row is the change in that effect after August 2015. Fines and their interaction with the post-2015 indicator are instrumented with three lags of ERA5 low cloud cover and their interactions with the indicator. Municipality and year-month fixed effects and weather controls included; standard errors clustered by municipality in parentheses.

7.2 Effect across different time exposure windows

Equation (15) uses a one-month exposure window. Deterrence may instead operate through learning: a farmer's decision to burn depends on the perceived risk of detection and punish-

ment, which is shaped by enforcement activity accumulated over time and diffused through social networks. A single month of exposure might be too short for that process. I therefore estimate how the effect changes with the length of the exposure window. For each window length $h \in \{1, 2, \ldots, 12\}$ months, I estimate the first stage

$$\bar{E}_{it}^{h} = \beta_h \, \overline{\text{DETERclouds}}_{it}^{h} + \boldsymbol{\gamma}_h \bar{\mathbf{X}}_{it}^{h} + \boldsymbol{\delta} \mathbf{X}_{it} + \alpha_i + \omega_t + \varepsilon_{it} \tag{19}$$

where

$$\bar{E}_{it}^{h} = \frac{1}{h} \sum_{s=1}^{h} E_{i,t-s} \qquad \text{average number of deforestation fines in } i \text{ in the past } h \text{ months,}$$

$$\overline{\text{DETERclouds}}_{it}^{h} = \frac{1}{h} \sum_{s=1}^{h} \text{DETERclouds}_{i,t-s} \qquad \text{average cloud cover in the past } h \text{ months,}$$

$$\bar{\mathbf{X}}_{it}^{h} = \frac{1}{h} \sum_{s=1}^{h} \mathbf{X}_{i,t-s} \qquad \text{rolling average of weather controls over the past } h \text{ months,}$$

and $\mathbf{X}_{it}$ are the contemporaneous weather controls. The matched rolling weather controls $\bar{\mathbf{X}}_{it}^{h}$ absorb collinearity between cloud cover and seasonal climate patterns over the enforcement window. For each h, the matched second stage is

$$\text{F}_{i,t+1} = \theta_h \hat{\bar{E}}_{it}^{h} + \boldsymbol{\gamma}_h \bar{\mathbf{X}}_{it}^{h} + \boldsymbol{\delta} \mathbf{X}_{it} + \psi_i + \lambda_t + \epsilon_{it} \tag{20}$$

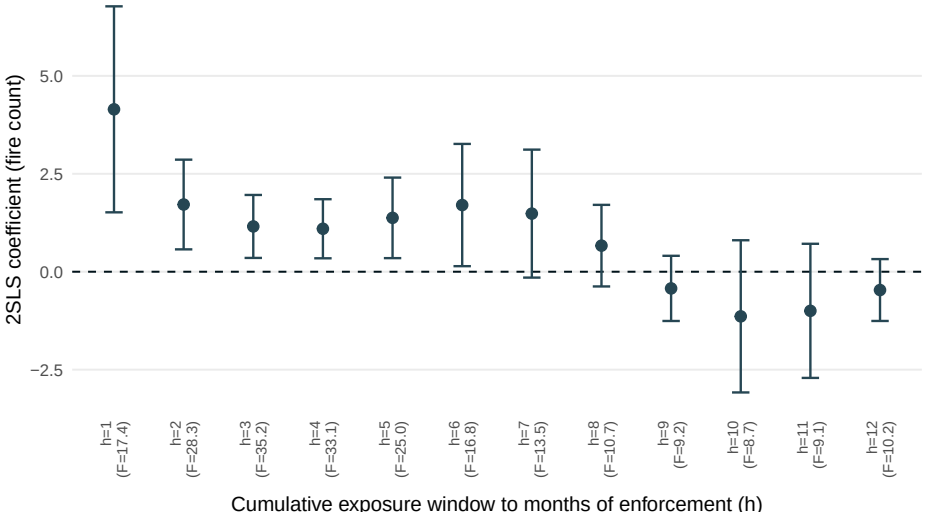
where $\hat{\bar{E}}_{it}^{h}$ is the h-month rolling average of fines instrumented by the rolling cloud cover

average. This yields a sequence of estimates $\{\hat{\theta}_h\}_{h=1}^{12}$ that traces how the effect of enforcement exposure on fire use changes with the window.

Figure 9 shows that the positive effect is concentrated in the short and medium run. The estimate is 4.15 for a one-month window ($p < 0.01$), falls to 1.16 for three months ($p < 0.01$) and is 1.70 for six months ($p < 0.05$). The effect is positive and significant for windows of up to six months, and remains positive but is not clearly distinguishable from zero at seven and eight months. For windows of nine months or more, the estimates are close to zero or negative and insignificant (-0.47 at twelve months, s.e. 0.40). The first-stage F-statistics also fall below 10 for several windows of nine months or more (Figure 9), so these long-window estimates are less precise.

The coefficients are not directly comparable across windows, since the regressor is a rolling average whose variation changes with h. The decay is nonetheless informative about timing. Fire use responds to enforcement within the first half-year and the response fades afterwards. This is consistent with the response being short-lived. It is also consistent with the general-deterrence channel in Section 3 becoming more important as enforcement accumulates, offsetting substitution toward fire. I cannot separate these explanations with the present design.

Figure 9: Effect of enforcement on fire ignitions across exposure windows



Note: Each point shows the IV coefficient (with 95% CI) on fire ignitions from a separate regression using a rolling average of deforestation fines over the prior h months, for $h = 1$ to 12 (Equation (20)). First-stage F-statistics for each window are reported on the x-axis. Municipality and year-month fixed effects and weather controls included.

8 Policy implications and conclusion

Environmental regulation and enforcement can induce unexpected behavioral responses. Understanding these responses is key to improving existing policies and designing better incentives to achieve conservation goals. To my knowledge, this paper presents the first evidence on how deforestation monitoring affects deforestation technology choice. I focus on the Brazilian Amazon, a central conservation area that has suffered from persistent forest fires even as conservation policy has drastically reduced deforestation rates since 2005.

I find evidence consistent with a strategic substitution from mechanical clearing toward fire-based clearing in municipalities more exposed to environmental law enforcement. Using cloud cover as an enforcement instrument, I find that a one-unit increase in the number of deforestation fines in a municipality-month is associated with 5 additional fire ignitions

in forest-covered areas the following month. The estimate is similar when using an alternative cloud-cover measure from ERA5 (about 7 ignitions). This result is consistent with my theoretical prediction that DETER created an enforcement gap between fire-based and mechanical clearing. The effect is positive and significant for exposure windows of up to six months and is robust to alternative specifications. Moreover, I find no evidence that the effect weakened after the 2015 upgrade of DETER, which improved its capacity to detect fire, although this test has limited power because fire detection remained weak after the upgrade. The substitution toward fire therefore does not appear to be only a temporary artifact of the system's early design. These findings point to an additional channel for the persistence of fires in the Amazon: a strategic response to deforestation monitoring and regulation.

My findings complement Assunção et al. (2023a), who show that DETER-induced enforcement reduced PRODES end-of-year deforestation. The two papers address different responses to the same mechanism: monitoring changes landowners' incentives at different margins. While Assunção et al. (2023a) estimate the effect on the margin observed by the DETER satellite, I focus on the response on the margin it observes poorly. Because the outcomes used to evaluate the Plan for the Prevention and Control of Deforestation in the Amazon (PPCDAm) are based on PRODES deforestation, part of the measured decline may reflect a shift toward technologies that these systems detect less well, rather than an equivalent reduction in forest loss. This suggests that existing PPCDAm assessments may overstate conservation achievements.

This work has first-order implications for designing enforcement against illegal fire use. First, monitoring systems should be evaluated on whether their detection capacity is balanced across clearing technologies, not only on their performance for the technology they detect best. Second, because the effect of enforcement on fire persists after the 2015 upgrade, improving the technical capacity to detect fire does not by itself appear sufficient to eliminate the incentive to substitute toward it; what matters is whether detection translates into on-

the-ground enforcement against fire-based clearing. Third, punishment of illegal fire use is a necessary complement to detection, since monitoring alone redirects behavior rather than eliminating it.

This paper leaves two questions for future work. The first is to quantify the externalities of the additional fire use, including its health costs. The second is to investigate the mechanisms behind the substitution using georeferenced enforcement data, including whether punishment deters fire use among neighboring landowners.

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Appendices

A Estimates: details and robustness

A.1 Clouds and smoke

A concern for the instrument is that smoke from fires could be classified as cloud, raising measured cloud cover wherever fires occur. Cloud cover would then be correlated with fire activity through a channel other than enforcement.

To address this, I test whether cloud cover rises with fire ignitions. I regress each cloud measure on fire ignitions in the same municipality-month, absorbing municipality and year-month fixed effects (Table A.10). Fires burn when the weather is dry and clear, which mechanically makes the correlation negative. For this reason, I add weather controls (columns 2 and 5). In the most demanding specification (columns 3 and 6), I also add last month's ignitions to allow smoke to persist, and replace the municipality fixed effects with municipality-by-calendar-month fixed effects, which absorb municipality-specific seasonality. The associations are small throughout, and their sign depends on the specification. In the most demanding specification, a one-standard-deviation increase in ignitions (about 12 fires) is associated with a change of -0.008 standard deviations in DETER cloud cover and -0.015 standard deviations in ERA5 cloud cover. The 95% confidence intervals lie at or below zero, whereas smoke contamination would produce a positive association. Combined exposure over the current and previous month is indistinguishable from zero for DETER and slightly negative for ERA5. Because year-month fixed effects absorb smoke episodes common to the whole Amazon, the test relies on differences across municipalities within a month. These estimates bound the net association between fires and measured cloud cover, but I cannot rule out small offsetting weather effects.

Table A.10: Cloud cover and fire ignitions

	DETER Cloud Cover			ERA5 Total Cloud Cover		
	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Fire ignitions (count)	-0.0006***	0.0001	-0.0003**	-0.0008***	0.0002***	-0.0003***
	(0.0002)	(9.12×10^{-5})	(0.0001)	(0.0002)	(4.99×10^{-5})	(6.35×10^{-5})
Fire ignitions$_{t-1}$			0.0002			8.22×10^{-5}***
			(9.48×10^{-5})			(2.75×10^{-5})
Weather controls	No	Yes	Yes	No	Yes	Yes
Sum: Fire$_t$ + Fire$_{t-1}$			-0.00013 (0.00018)			-0.00020*** (0.00005)
Fixed-effects						
Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Municipality × calendar-month FE	No	No	Yes	No	No	Yes
Municipality FE	Yes	Yes	No	Yes	Yes	No
Fit statistics						
R^2	0.61638	0.64353	0.69613	0.84397	0.94156	0.96572
Within R^2	0.00062	0.07135	0.02564	0.00842	0.62864	0.35268
Observations	104,326	104,326	104,326	104,326	104,326	104,326

Clustered (muni_code) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

B Tables

Table B.1: Municipality-level monthly variables: summary statistics

Table B.2: Summary Statistics

Variable	Mean	SD	Min	Max
Fire ignitions (count)	0.720	4.859	0.000	408.000
Deforestation fines	0.255	2.185	0.000	175.000
ERA5 average cloud cover	0.638	0.226	0.043	0.974
DETER cloud cover	0.394	0.433	0.000	1.000
Rainfall (mm)	159.249	134.575	0.000	854.851
Mean temperature	26.257	1.536	20.033	32.755
Relative humidity	75.358	12.964	26.368	93.262
Evapotranspiration (mm/day)	3.146	0.932	0.538	5.235
Soil moisture, 0–100 cm	0.347	0.096	0.067	0.500
N = 164,649 municipality-month observations.				

Table B.3: IV First stage results

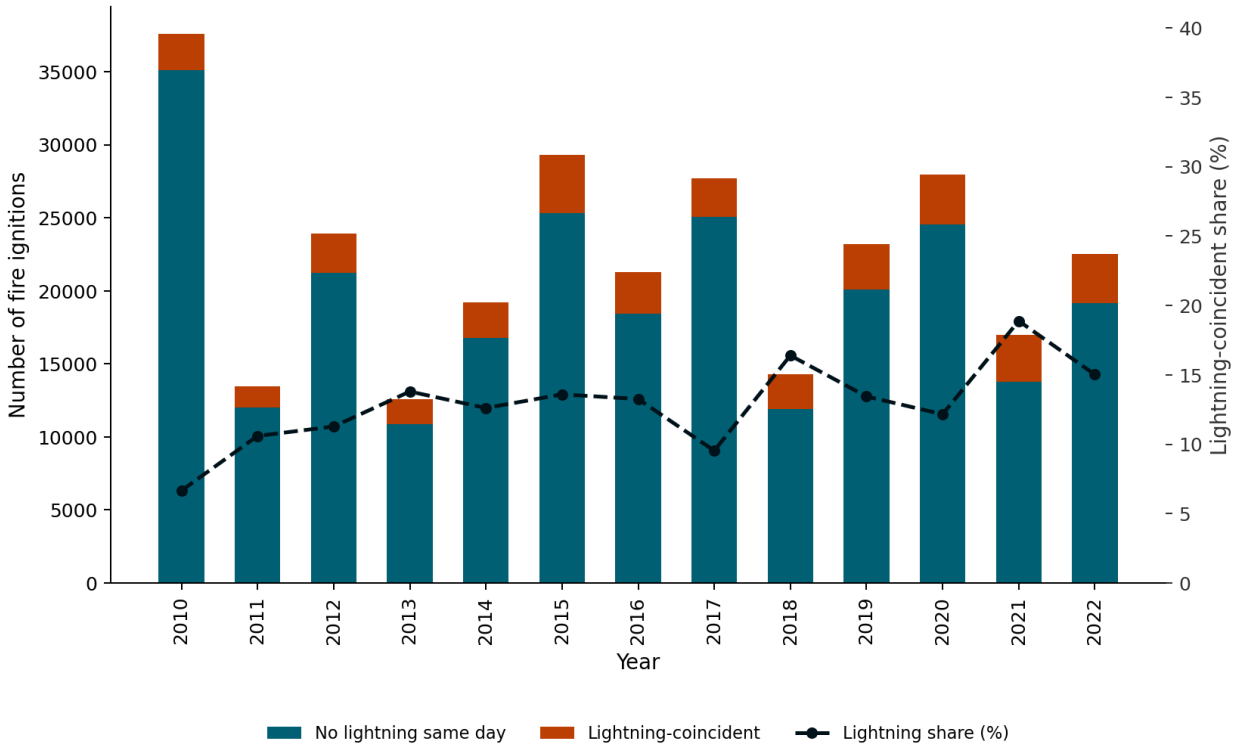
Dependent Variable:	Enforcement (deforestation fines$_t$)	
		IV
Model:	(1)	(2)
Variables		
Deter cloud$_t$	-0.0338	
	(0.0268)	
Deter cloud$_{t-1}$	-0.0586***	
	(0.0225)	
Deter cloud$_{t-2}$	-0.1016***	
	(0.0232)	
Rainfall (mm)	0.0002*	7.4×10^{-5}
	(0.0001)	(9.5×10^{-5})
Temperature (°C)	-0.0752***	-0.0111
	(0.0180)	(0.0105)
Evaporation (mm/day)	0.1362***	0.0626***
	(0.0292)	(0.0182)
Soil moisture (0–100 cm)	0.1012	-0.0186
	(0.3952)	(0.3995)
Relative humidity	-0.0172***	
	(0.0034)	
ERA5 cloud$_t$		-0.4650***
		(0.1744)
ERA5 cloud$_{t-1}$		-0.3614**
		(0.1640)
ERA5 cloud$_{t-2}$		-0.2369*
		(0.1324)
Fixed-effects		
Municipality FE	Yes	Yes
Year-Month FE	Yes	Yes
Fit statistics		
F-test (1st stage)	10.479	15.124
Observations	96,538	106,674

Clustered (Municipality) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Note: First-stage estimate from the main estimate (Table 3) of the effect of environmental enforcement on land-clearing patterns. Estimates consider the 2007-2019 period.

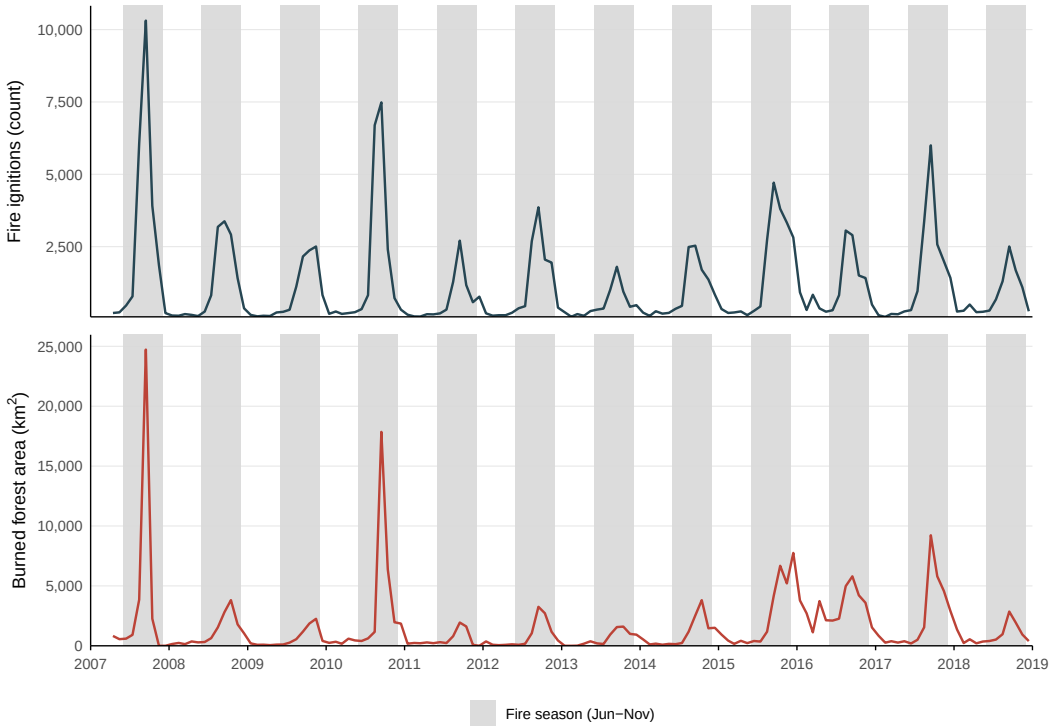
C Figures

Figure C.1: Lightning-coincident versus likely human-caused fire ignitions in the Amazon, 2010–2022



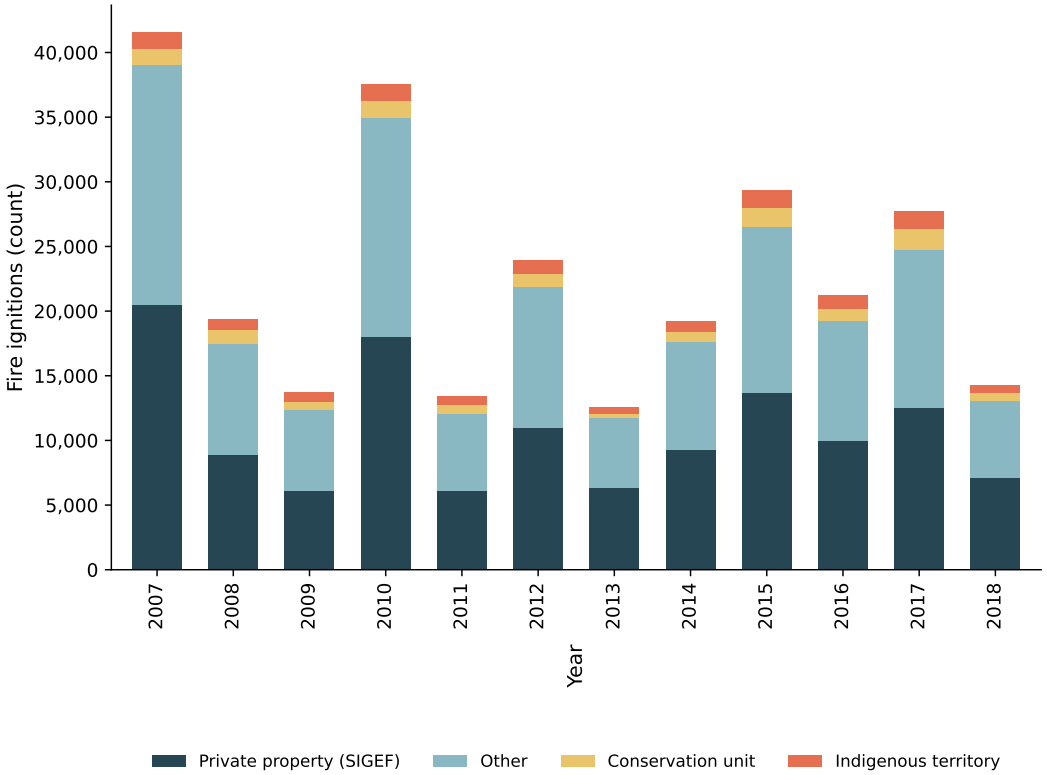
Note: orange) and those with no same-day lightning (teal); the dashed line (right axis) shows the lightning-coincident share. A fire is classified as lightning-coincident if the WWLLN Global Lightning Climatology records a positive stroke density in the same 0.5° grid cell on the same calendar day as the ignition. Because WWLLN detects only about 30% of global lightning strokes, the lightning-coincident share should be treated as a lower bound.

Figure C.2: Fire ignitions and burned forest areas, 2007–2018



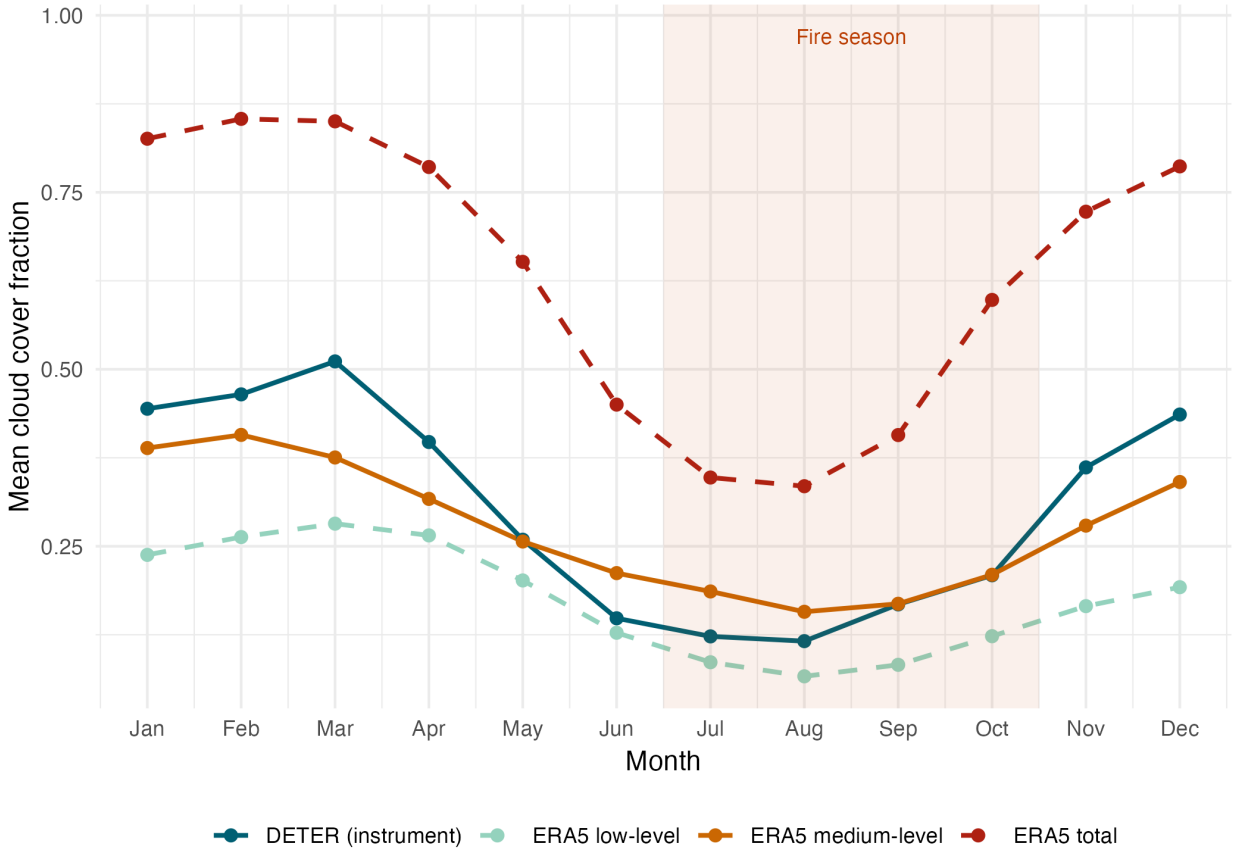
Note: Monthly in the Legal Amazon. The top panel shows fire ignitions, identified from satellite fire data with the Balboni et al. (2023) fire-event algorithm applied to MODIS. The bottom panel shows burned forest area in km², from MapBiomas Fire, counting only burned pixels on forest-covered areas. Grey bands mark the fire season (June–November).

Figure C.3: Fire ignitions by tenure, 2007–2018



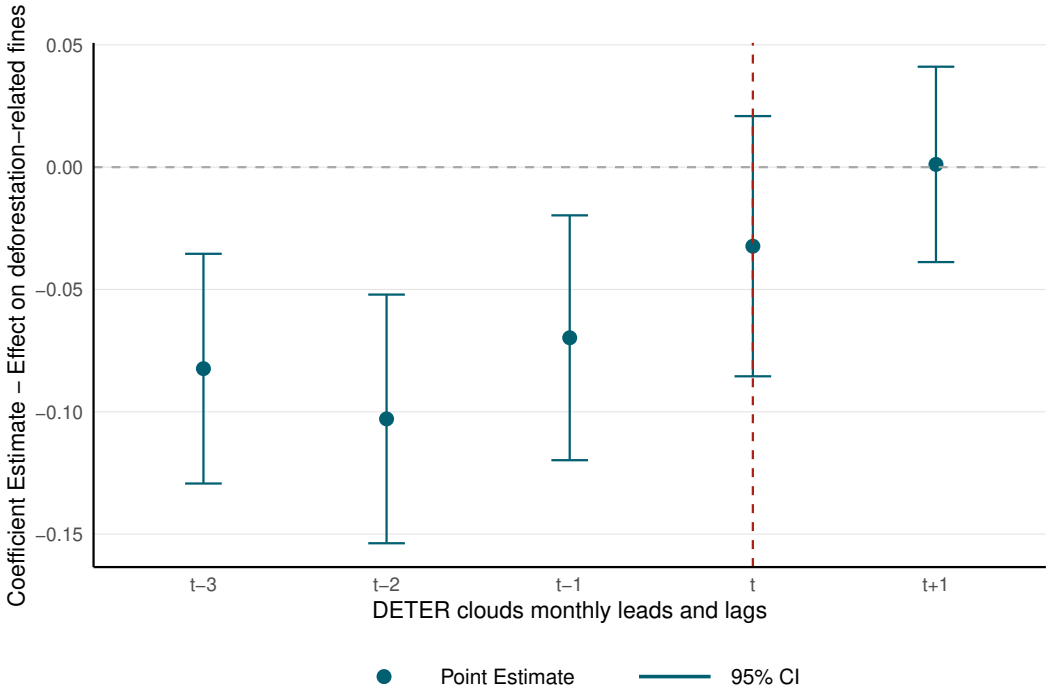
Note: Each bar shows the number of fire ignitions in a year by land tenure category at the ignition point within the Legal Amazon. Tenure is assigned by overlaying ignition points on SIGEF-registered private properties, conservation units, and indigenous territories in the Amazon biome. When a point falls in more than one layer, it is assigned in the order: indigenous territory, conservation unit, private property. "Other" includes ignitions outside all three layers, such as undesignated public land and uncertified properties. Layers are static boundaries as in 2024, and are applied to all years.

Figure C.4: Comparison of cloud cover products across the seasonal cycle



Note: Mean monthly cloud cover fraction, pooled across years, for four measures: DETER's operational instrument, ERA5 low-level cloud cover, ERA5 medium-level cloud cover, and ERA5 total cloud cover. The shaded region marks the fire season (roughly June–October). All four measures decline heading into the dry/fire season and rise again afterward; DETER's cloud cover tracks the ERA5 low-level measure most closely among the ERA5 series.

Figure C.5: Robustness check: cloud cover instrument, leads and lags



Note: Coefficient estimates (with 95% CI) from separate regressions of deforestation-related fines on ERA5 cloud cover at monthly leads and lags from t-3 to t+2, controlling for municipality and year-month fixed effects. Cloud cover from t-3 through t is negatively and significantly associated with fines; cloud cover at t+1 and t+2 shows no significant association, supporting the exclusion restriction.